

Fourier transforms

This course aims to provide an introduction to *Fourier transform* techniques – powerful tools for analysis of the properties of a wide range of signals and systems.

1. INTRODUCTION

1.1. TRANSFORMS IN ENGINEERING

Throughout science and engineering, choosing to transform the way in which a problem is viewed is a common technique for making data, signals or models easier to manipulate or understand. Logarithms, for example, allow the representation of data over a wide range of scales, can be used to turn the process of multiplying two numbers into that of simple addition, and can be applied to entire equations or to graphs, resulting in alternative representations such as Bode Plots which can, for example, make it easier to establish the dominant power-law behaviour of a system over a wide range of frequencies. Likewise, Laplace transforms can be applied to differential equation models of continuous-time dynamic systems in order to turn the processes of differentiation and integration into that of simple multiplication or division by the complex frequency variable s , and greatly assist in handling systems design since it is correspondingly much easier to break large-scale systems into series and parallel structures involving a set of simpler, interconnected subsystems. Also, beyond the Laplace transform there is a further tool, called the Z-transform, which is appropriate for working with digital, sampled-data signals and systems, and which is used heavily for digital control theory and digital signal processing.

There are, in fact, a range of transform techniques in use for specific engineering purposes, the most common of which are the Laplace transform, the Z-transform, and the Fourier transform (named after Jean-Baptiste Joseph Fourier, 1768-1830) which is mainly applied to signals and typically turns the viewpoint of the signal from the time-domain to the frequency domain. Depending on the application area, you might also encounter the wavelet transform, the Hilbert transform, etc.

1.2. REVIEW - THE LAPLACE TRANSFORM

Recall that the *single-sided Laplace transform* is defined by the expression

$$F(s) = L[f(t)] \triangleq \int_0^{\infty} f(t) e^{-st} dt \quad (1)$$

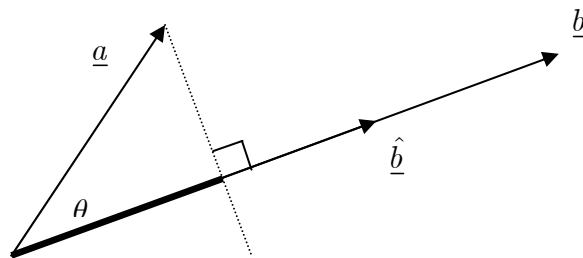
where it is assumed that $f(t) = 0$ for $t < 0$. The transform changes the viewpoint from that of a (real) time-domain function $f(t)$ to that of a (complex) s -domain function $F(s)$ where $s = \sigma + j\omega$ is known as the *complex frequency variable* or the *Laplace variable*.

Also, the specific lower limit allows for the concept of a signal that starts at $t = 0$, or at some delayed time.

In particular, an analogy was drawn with the basic operation of a vector dot-product, which is based on the physical operation of projecting one vector onto another.

$$\underline{a} \cdot \underline{b} = ab \cos \theta \quad (2)$$

so that if $\hat{\underline{b}}$ is a unit vector $\underline{a} \cdot \hat{\underline{b}}$ is the projection of $\underline{a}$ onto the direction $\hat{\underline{b}}$ - producing a pure length (scalar number) which can be interpreted as the amount of $\underline{a}$ in the direction $\hat{\underline{b}}$:



In Cartesian coordinates, the dot product takes the form

$$\underline{a} \cdot \underline{b} = a_x b_x + a_y b_y + a_z b_z$$

so that the basic process (apart from a normalizing factor) is that of element-by-element multiplication, followed by a summation, producing a scalar result. The dot product is a basic example of a larger set of mathematical processes called *inner products*, of which the Laplace transform is also an example. Although working with continuous functions rather than vectors, the same broad interpretation

applies; for a given, fixed value of s , the Laplace transform is obtained by multiplying the function $f(t)$ by e^{-st} (time-domain modulation), and then integrating, producing a scalar result. Broadly speaking, this scalar result can be thought of as a measure of the amount of $f(t)$ in the "direction" e^{-st} .

1.3. ORTHOGONALITY AND NORMALIZATION

Pursuing the idea of projection, one important use of the dot product when working with simple vectors was to establish whether two vectors were orthogonal. Since

$$\underline{a} \cdot \underline{b} = ab \cos(\theta) = a_x b_x + a_y b_y + a_z b_z$$

for non-zero vectors, $\underline{a} \cdot \underline{b} = 0 \Leftrightarrow \underline{a} \perp \underline{b}$ (since $\theta = \frac{\pi}{2}$).

For example, if

$$\underline{a} = \begin{pmatrix} 1 \\ -1 \\ 5 \end{pmatrix} \quad \text{and} \quad \underline{b} = \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix}$$

$$\text{then } \underline{a} \cdot \underline{b} = \begin{pmatrix} 1 \\ -1 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix} = (1)(-2) + (-1)(3) + (5)(1) = 0$$

Therefore $\cos \theta = 0$, and so the vectors are orthogonal.

Further, it follows that

$$\underline{a} \cdot \underline{a} = a^2 \quad (\text{since } \theta = 0)$$

which allows the normalization of a vector to unit length, since

$$\hat{\underline{a}} = \frac{\underline{a}}{|\underline{a}|}$$

These concepts are valid for inner products as well. A common notation for the inner product of two (potentially complex) functions would be

$$\langle f, g \rangle = \int_{x_1}^{x_2} \overline{f(x)} g(x) dx \quad (3)$$

where $\overline{f(x)}$ is the complex conjugate of $f(x)$ and the integration is

carried out over a specified *interval* $[x_1, x_2]$.

As with the vector dot product $\underline{a} \cdot \underline{b}$, two functions are said to be orthogonal over the interval $[x_1, x_2]$ if

$$\langle f, g \rangle = \int_{x_1}^{x_2} \overline{f(x)} g(x) dx = 0 \quad (4)$$

and, as with the special case $\underline{a} \cdot \underline{a} = |\underline{a}|^2 = a^2$ for vectors, in particular,

$$\langle f, f \rangle = \int_{x_1}^{x_2} \overline{f(x)} f(x) dx \quad (5)$$

must be greater than zero if the function is non-zero anywhere within the interval of interest. This integral gives a measure of the "size" of the function and, as with vectors, can be used to normalize a function. In particular, for any function $f(x)$, it is possible to define a scaled version

$$\hat{f}(x) = \frac{f(x)}{\sqrt{\int_{x_1}^{x_2} \overline{f(x)} f(x) dx}} \quad (6)$$

which is of unit "size" in the sense that its inner product with itself would be one:

$$\langle \hat{f}, \hat{f} \rangle = 1 \quad (7)$$

Orthogonal functions are a particularly convenient tool in science and engineering; they play a similar role to orthogonal axes in simple Cartesian 2- and 3-dimensional space. In short, they are *independent* of each other; in 2D on the Earth's surface, you know that if you move 3m due north, you have not moved at all in the east/west direction. Likewise in 3D Cartesian space, the position vector

$$\underline{r} = x\underline{\hat{i}} + y\underline{\hat{j}} + z\underline{\hat{k}} \quad (8)$$

represents three independent steps in three orthogonal directions along XYZ axes defined by three *orthonormal basis vectors* $\underline{\hat{i}}$, $\underline{\hat{j}}$ and $\underline{\hat{k}}$.

Orthonormal basis vectors can be used to provide an unambiguous way to describe the vector from the origin to a point in n -dimensional space as the sum of a series of independent, orthogonal steps of different sizes. Likewise a set of orthonormal basis functions can be used to construct an alternative representation of a given function as

the sum of a series of weighted, independent, orthogonal functions. The concept of using basis functions is not new; the infinite series expansion of $\exp(x)$ can be written as

$$\exp(x) \equiv 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \cdots + \frac{1}{k!}x^k + \cdots \quad \forall x \quad (9)$$

in terms of the basis function $1, x, x^2, x^3$, etc., but these functions are neither orthogonal to each other nor normalized over any finite interval. Moreover, although the representation in terms of basis functions is useful, it doesn't offer any particular physical insights, or relate to parameters that we might wish to measure.

1.4. SINUSOIDAL BASIS FUNCTIONS

Some potential basis functions that do offer obvious practical insights are sinusoidal functions. Each sinusoid is defined precisely by just three parameters; frequency, amplitude and phase. Further, restricting the functions to just pure cosine and sine signals fixes the phase parameter, so that the only variables are frequency and amplitude. This means that sinusoids offer the possibility of providing a set of basis functions that can be used to describe a signal in an interpretable fashion – the relative amounts of different frequency components that it might contain.

First, it is necessary to establish some basic properties of sines and cosines.

Here, we will most often be working with continuous-time signals, so the theory will be developed with one-dimensional integrals with respect to time. But these basic ideas can also be extended to multiple dimensions and, in particular, 2D image processing, in which case the key variables may be distances (the relative positions of pixels in an image) rather than time.

Cosine functions

First consider the specific integral

$$\int_{-T/2}^{T/2} \cos(m\omega_0 t) \cos(n\omega_0 t) dt \quad (10)$$

where $\omega_0 = \frac{2\pi}{T}$ and where m and n are positive integers.

This has the form of an inner product over a time interval $\left[-\frac{T}{2}, \frac{T}{2}\right]$ where the two real functions involved are $f(t) = \cos(m\omega_0 t)$ and $g(t) = \cos(n\omega_0 t)$. Since both functions are real, there is no need for the complex conjugate defined in equation (4).

Each function represents an integer number of complete cosine cycles over the specified time interval, while m and n simply count the number of cycles in each case. To solve the integral, we can use a standard trigonometric identity:

$$\cos(\alpha)\cos(\beta) \equiv \frac{1}{2}[\cos(\alpha + \beta) + \cos(\alpha - \beta)] \quad (11)$$

First, if $m \neq n$ we can write

$$\begin{aligned} & \int_{-T/2}^{T/2} \cos(m\omega_0 t) \cos(n\omega_0 t) dt \\ &= \frac{1}{2} \int_{-T/2}^{T/2} (\cos((m+n)\omega_0 t) + \cos((m-n)\omega_0 t)) dt \\ &= \frac{1}{2\omega_0} \left[\frac{1}{(m+n)} \sin((m+n)\omega_0 t) + \frac{1}{(m-n)} \sin((m-n)\omega_0 t) \right]_{-T/2}^{T/2} \end{aligned}$$

so that

$$\int_{-T/2}^{T/2} \cos(m\omega_0 t) \cos(n\omega_0 t) dt = 0 \quad \text{for } m \neq n \quad (12)$$

since $\omega_0 = \frac{2\pi}{T}$, and $\sin((m+n)2\pi) = \sin((m-n)2\pi) = 0$ for all integer values of m and n .

Hence the functions $\cos(m\omega_0 t)$ and $\cos(n\omega_0 t)$ are orthogonal to each other over a time interval $\left[-\frac{T}{2}, \frac{T}{2}\right]$ for all pairs of positive integers m and n where $m \neq n$.

Also, in the particular case where $m = n$ we can write

$$\begin{aligned} & \int_{-T/2}^{T/2} \cos(m\omega_0 t) \cos(m\omega_0 t) dt \\ &= \int_{-T/2}^{T/2} \cos^2(m\omega_0 t) dt = \frac{1}{2} \int_{-T/2}^{T/2} (\cos(2m\omega_0 t) + 1) dt \quad (13) \\ &= \frac{1}{2} \left[\frac{1}{2m\omega_0} \sin(2m\omega_0 t) + t \right]_{-T/2}^{T/2} = \frac{1}{2} \left(\frac{T}{2} - -\frac{T}{2} \right) = \frac{T}{2} \end{aligned}$$

Since $\cos^2(\alpha) \equiv \frac{1}{2}[\cos(2\alpha) + 1]$.

Finally, in the special case where either m or n (but not both) are zero, we also get

$$\int_{-T/2}^{T/2} \cos(m\omega_0 t) 1 dt = \int_{-T/2}^{T/2} 1 \cos(n\omega_0 t) dt = 0 \quad (14)$$

and if both $m = n = 0$ we can write

$$\int_{-T/2}^{T/2} 1 dt = [t]_{-T/2}^{T/2} = T \quad (15)$$

These possibilities encompass all possible pairs of non-negative integer values of m and n and show that the (infinite set of) pure cosine functions $\cos(m\omega_0 t)$ and $\cos(n\omega_0 t)$ are orthogonal for all non-negative values of m and n over a time interval of length T .

NOTE These results remain valid for any time interval of duration T . Here, the interval was chosen to be $[-\frac{T}{2}, \frac{T}{2}]$, but we could equally have used $[0, T]$, for example. Many texts indicate this by using the general notation

$$\int_{\langle T \rangle} \cos(m\omega_0 t) \cos(n\omega_0 t) dt$$

where the $\langle T \rangle$ merely indicates a time interval of length T .

In short:

$$\int_{-T/2}^{T/2} \cos(m\omega_0 t) \cos(n\omega_0 t) dt = \begin{cases} \frac{T}{2} & n = m \neq 0 \\ T & n = m = 0 \\ 0 & n \neq m \end{cases}$$

Sine functions

Similarly, considering pure sine functions:

$$\int_{-T/2}^{T/2} \sin(m\omega_0 t) \sin(n\omega_0 t) dt \quad (16)$$

where $\omega_0 = \frac{2\pi}{T}$ and where m and n are positive integers, and using a different trigonometric identity:

$$\sin(\alpha) \sin(\beta) \equiv \frac{1}{2} [-\cos(\alpha + \beta) + \cos(\alpha - \beta)] \quad (17)$$

Again, if $m \neq n$ we can write

$$\begin{aligned} & \int_{-T/2}^{T/2} \sin(m\omega_0 t) \sin(n\omega_0 t) dt \\ &= \frac{1}{2} \int_{-T/2}^{T/2} (-\cos((m+n)\omega_0 t) + \cos((m-n)\omega_0 t)) dt \\ &= \frac{1}{2\omega_0} \left[-\frac{1}{(m+n)} \sin((m+n)\omega_0 t) + \frac{1}{(m-n)} \sin((m-n)\omega_0 t) \right]_{-T/2}^{T/2} \end{aligned}$$

so that putting the limits in shows that

$$\int_{-T/2}^{T/2} \sin(m\omega_0 t) \sin(n\omega_0 t) dt = 0 \quad \text{for } m \neq n \quad (18)$$

Also, for two equal positive integers $m = n$ we can write

$$\begin{aligned} & \int_{-T/2}^{T/2} \sin(m\omega_0 t) \sin(m\omega_0 t) dt \\ &= \int_{-T/2}^{T/2} \sin^2(m\omega_0 t) dt = \frac{1}{2} \int_{-T/2}^{T/2} (1 - \cos(2m\omega_0 t)) dt \\ &= \frac{1}{2} \left[t - \frac{1}{2m\omega_0} \sin(2m\omega_0 t) \right]_{-T/2}^{T/2} = \frac{1}{2} \left(\frac{T}{2} - -\frac{T}{2} \right) = \frac{T}{2} \end{aligned} \quad (19)$$

And finally it is obvious that

$$\int_{-T/2}^{T/2} \sin(m\omega_0 t) \sin(n\omega_0 t) dt = 0 \quad (20)$$

if either $m = 0$ or $n = 0$ (or both) are zero.

Again, these possibilities encompass all possible pairs of non-negative integer values of m and n and show that the (infinite set of) pure sine functions $\sin(m\omega_0 t)$ and $\sin(n\omega_0 t)$ are orthogonal for all non-negative values of m and n over a time interval of length T .

In short:

$$\int_{-T/2}^{T/2} \sin(m\omega_0 t) \sin(n\omega_0 t) dt = \begin{cases} \frac{T}{2} & n = m \neq 0 \\ 0 & n = m = 0 \\ 0 & n \neq m \end{cases}$$

Mixed cosine and sine functions

Finally, consider a product of pure sine and cosine functions over a time interval of length T :

$$\int_{-T/2}^{T/2} \sin(m\omega_0 t) \cos(n\omega_0 t) dt \quad (21)$$

where $\omega_0 = \frac{2\pi}{T}$ and where m and n are positive integers. Using yet another trigonometric identity:

$$\sin(\alpha) \cos(\beta) \equiv \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)] \quad (22)$$

if $m \neq n$ we can write

$$\begin{aligned} & \int_{-T/2}^{T/2} \sin(m\omega_0 t) \cos(n\omega_0 t) dt \\ &= \frac{1}{2} \int_{-T/2}^{T/2} (\sin((m+n)\omega_0 t) + \sin((m-n)\omega_0 t)) dt \\ &= \frac{1}{2\omega_0} \left[-\frac{1}{(m+n)} \cos((m+n)\omega_0 t) - \frac{1}{(m-n)} \cos((m-n)\omega_0 t) \right]_{-T/2}^{T/2} \end{aligned}$$

so that putting the limits in shows that

$$\int_{-T/2}^{T/2} \sin(m\omega_0 t) \cos(n\omega_0 t) dt = 0 \quad \text{for } m \neq n \quad (23)$$

This time, for the special case of two equal positive integers $m = n$ we can write

$$\begin{aligned} & \int_{-T/2}^{T/2} \sin(m\omega_0 t) \cos(m\omega_0 t) dt \\ &= \frac{1}{2} \int_{-T/2}^{T/2} \sin(2m\omega_0 t) dt = \frac{1}{4\omega_0} [-\cos(2m\omega_0 t)]_{-T/2}^{T/2} = 0 \end{aligned}$$

Also the integral vanishes if either (or both) $m = 0$ or $n = 0$. So that

$$\int_{-T/2}^{T/2} \sin(m\omega_0 t) \cos(n\omega_0 t) dt = 0 \quad (24)$$

for all non-negative integers m and n .

In short, $\sin(m\omega_0 t)$ and $\cos(n\omega_0 t)$ are orthogonal for all non-negative values of m and n .

Summary

Overall, we have that facts that over a time interval of length T all sine and cosine functions of the form $\sin(m\omega_0 t)$ and $\cos(n\omega_0 t)$ for non-negative integers m and n , and where $\omega_0 = \frac{2\pi}{T}$, form an orthogonal set:

$$\int_{-T/2}^{T/2} \cos(m\omega_0 t) \cos(n\omega_0 t) dt = 0 \quad \text{for } m \neq n$$

$$\int_{-T/2}^{T/2} \sin(m\omega_0 t) \sin(n\omega_0 t) dt = 0 \quad \text{for } m \neq n$$

$$\int_{-T/2}^{T/2} \sin(m\omega_0 t) \cos(n\omega_0 t) dt = 0 \quad \text{for all values of } m \text{ and } n.$$

and

$$\int_{-T/2}^{T/2} \cos(m\omega_0 t) \cos(n\omega_0 t) dt = \frac{T}{2} \quad \text{for } m = n \neq 0$$

$$\int_{-T/2}^{T/2} \cos(m\omega_0 t) \cos(m\omega_0 t) dt = T \quad \text{for } m = n = 0$$

$$\int_{-T/2}^{T/2} \sin(m\omega_0 t) \sin(n\omega_0 t) dt = \frac{T}{2} \quad \text{for } m = n \neq 0$$

This means that the sinusoids can form a useful set of orthogonal, independent basis functions for representing signals unambiguously in terms of their frequency content. Methods based around this idea are often referred to as Fourier Analysis (named after Jean-Baptiste Joseph Fourier, 1768-1830).

Many textbooks use the notation $\Phi_m(t)$ to represent the specific elements of a set of basis functions, where n is merely a positive integer which counts through the elements of the set. Also, many texts use the notation $\Phi_m^*(t)$ to represent the complex conjugate of a function.

In this notation, a set of such basis functions, are said to be orthogonal over the time interval (a, b) if

$$\int_a^b \Phi_m(t) \Phi_n^*(t) dt = \begin{cases} E_n & n = m \\ 0 & n \neq m \end{cases}$$

where $E_n = \int_a^b |\Phi_m(t)|^2 dt$

2. FOURIER METHODS

The results in the previous section can be applied to provide a frequency-domain representation of a signal, but only over a restricted time interval of length T . This is the basis of a particular method called *Fourier series*.

Textbooks refer to Fourier methods, or Fourier analysis or even just "*the Fourier Transform*", which can be confusing and misleading. Although there is a Fourier transform, it is in fact just one very specific member of a family of transforms which together form a group of closely-related techniques and which can be applied to different signal types for different purposes. What all Fourier techniques have in common is the idea that signals can be characterized by their frequency content. In effect, the signal can be viewed as being constructed from a weighted sum of perfect orthogonal sinusoidal signals of differing frequencies; the problem then becomes how to evaluate the amount of any given frequency within the signal of interest.

The idea of representing a signal or function as a weighted sum of basis functions is not new - as mentioned above, functions can be represented using a polynomial basis set, and such an approach may well be the best one to use for purposes such as linearization of a complicated system or for constructing short-range approximations (interpolation or extrapolation) of a signal.

The key assumption underlying all Fourier methods is that it is reasonable or meaningful to represent a signal/function in terms of a sinusoidal basis set; but as with polynomial expansions, this may not always be the best approach - the appropriateness of the tools depends on the signal properties and the application context. For example, the frequency content of a signal may change over time and, in the cases of speech or music, often does so quite rapidly.

Periodic functions

Starting with the idea that sinusoids can form an orthogonal set of functions over a restricted time interval of length T , *Fourier series* are the appropriate tool to be applied to any chosen single period of a *periodic* function of period T , such as those illustrated below. These functions all have the same repetition period and are assumed to

repeat indefinitely.

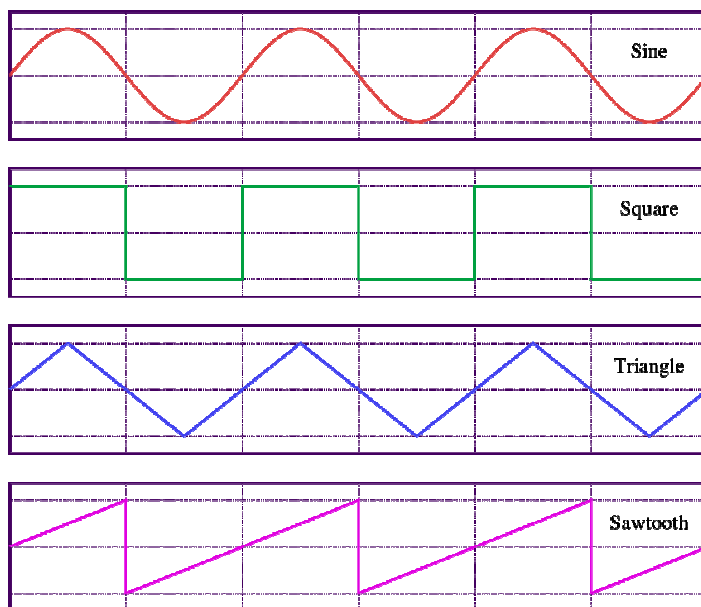


Figure 1 Examples of periodic functions (source: Wikipedia)

Note that there is a major difference between the concept of period and frequency. These four functions all have the same period (say T), but they do **not** have the same frequency. Only the sine wave has a single, specific well-defined frequency - the others can be viewed as containing multiple (in theory, an infinite number) of frequencies.

Aperiodic functions are simply those that do not display such regular (in theory, infinite) repeated patterns, such as any experimental signal, which is of course subject to random noise, and the three examples illustrated below.

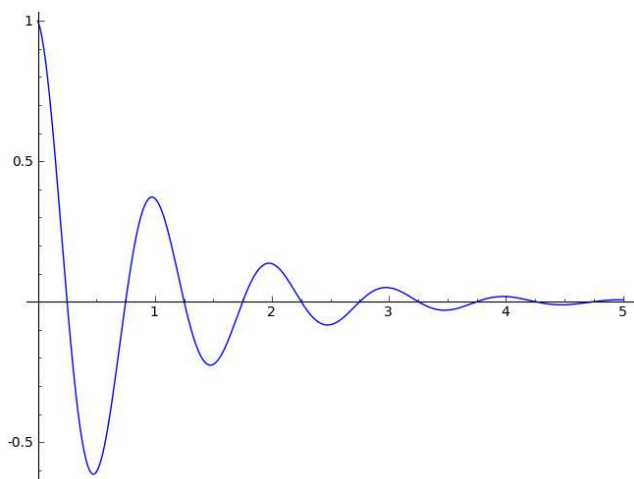


Figure 2 An aperiodic function of the form $e^{-\sigma t} \cos(2\pi\omega_0 t)$

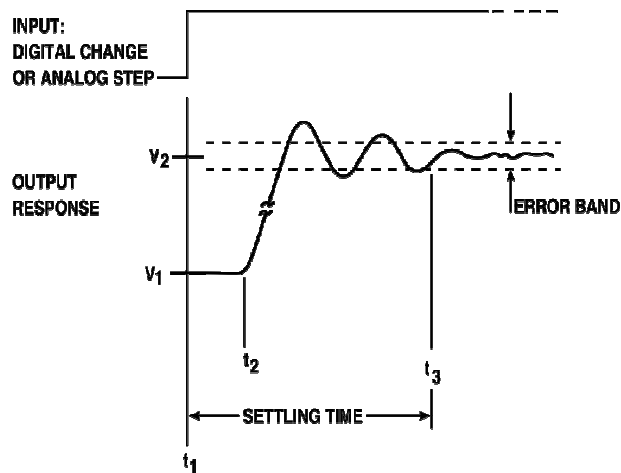


Figure 3 A typical aperiodic step response function, illustrating overshoot and settling (source: Wikipedia)

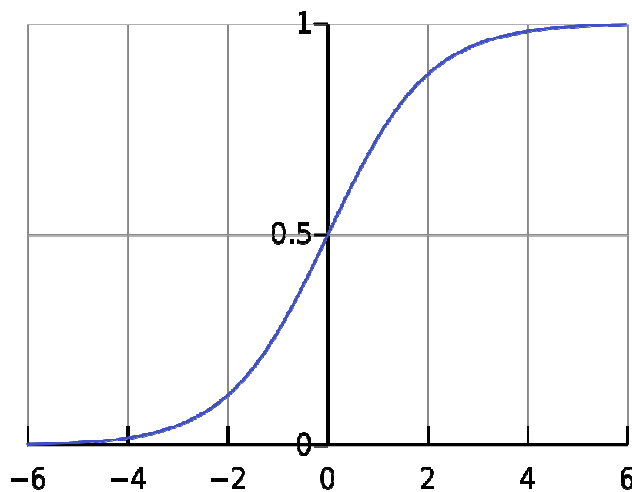


Figure 4 Another example of an aperiodic function is the logistic function. This is an S-shaped "sigmoid" curve with many applications. It is commonly used to represent systems with an approximately linear response for smaller levels of input but which "saturate" for larger levels so that here, for example, the outputs are very nearly constant (0 and 1) for inputs below -6 and above 6 respectively.

2.1. SINE-COSINE FOURIER SERIES

In general for a periodic function of period T , Fourier series represent a single period of a *piecewise continuously differentiable periodic function*, $x(t)$, as an infinite series of sine and cosine functions of specific frequencies:

$$x(t) = a_0 + \sum_{n=1}^{n=\infty} (a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)) \quad (25)$$

where $\omega_0 = \frac{2\pi}{T}$ and the Fourier coefficients are given by

$$a_0 = \frac{1}{T} \int_{\langle T \rangle} x(t) dt \quad (26)$$

$$a_n = \frac{2}{T} \int_{\langle T \rangle} x(t) \cos(n\omega_0 t) dt \quad (27)$$

$$b_n = \frac{2}{T} \int_{\langle T \rangle} x(t) \sin(n\omega_0 t) dt \quad (28)$$

Note that the specific set of frequencies used for the Fourier expansion, $n\omega_0 = \frac{2\pi n}{T}$, depend entirely on the periodicity of the function - we do not choose the frequencies.

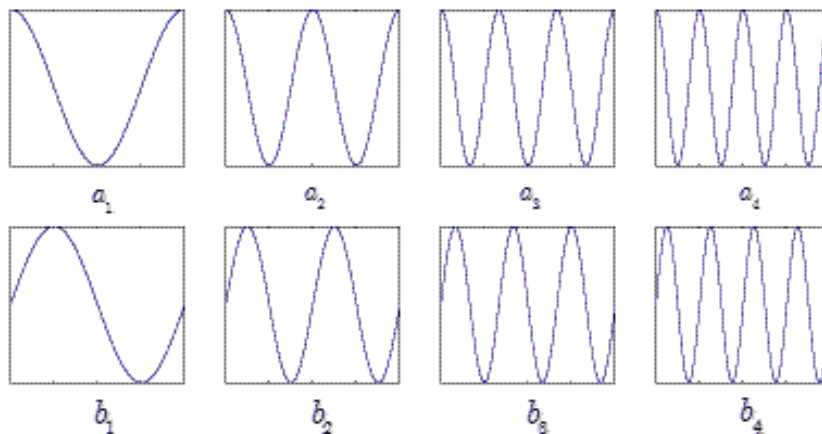


Figure 5 The cosine and sine basis functions of frequencies $\omega_0, 2\omega_0, 3\omega_0$, and $4\omega_0$.

The sinusoidal basis functions for a single period are shown above. the a_n and b_n coefficients control the relative amounts of each function in the infinite sum defined in equation (25). The a_0 coefficient is a special case - instead of a sinusoidal function, it reflects the average value of the signal over the entire period - a constant *offset*.

The values of the coefficients can be obtained by simply multiplying the series expansion of $x(t)$

$$x(t) = a_0 + \sum_{n=1}^{n=\infty} (a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t))$$

by, any basis function, and then integrating over the period.

Remember that the series counter is just a dummy variable and can be renamed as necessary.. So that, for example, to establish the value of the coefficient a_n we evaluate

$$\begin{aligned} & \int_{\langle T \rangle} x(t) \cos(n\omega_0 t) dt \\ &= \int_{\langle T \rangle} \left(a_0 + \sum_{k=1}^{k=\infty} (a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)) \right) \cos(n\omega_0 t) dt \\ &= \int_{\langle T \rangle} a_n \cos^2(n\omega_0 t) dt = \int_{\langle T \rangle} a_n \frac{1}{2} (1 + \cos(2n\omega_0 t)) dt = \frac{T}{2} a_n \end{aligned}$$

since every basis function is normal to $\cos(n\omega_0 t)$ except $\cos(n\omega_0 t)$.

Hence

$$a_n = \frac{2}{T} \int_{\langle T \rangle} x(t) \cos(n\omega_0 t) dt$$

The average value of the signal over the period is

$$\begin{aligned} & \int_{\langle T \rangle} x(t) dt = \int_{\langle T \rangle} \left(a_0 + \sum_{k=1}^{k=\infty} (a_k \cos(k\omega_0 t) + b_k \sin(k\omega_0 t)) \right) dt \\ &= \int_{\langle T \rangle} a_0 dt = T a_0 \end{aligned}$$

so that

$$a_0 = \frac{1}{T} \int_{\langle T \rangle} x(t) dt$$

Similarly

$$b_n = \frac{2}{T} \int_{\langle T \rangle} x(t) \sin(n\omega_0 t) dt$$

With these particular values of the *Fourier coefficients* the expression

$$x(t) = a_0 + \sum_{n=1}^{n=\infty} (a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t))$$

defines the *sine-cosine form* of the *Fourier series expansion* of $x(t)$.

Other forms are used as well, and are discussed below.

If $x(t)$ is a continuous function, the Fourier series expansion will generally converge to the value of $x(t)$ at any given time. If $x(t)$ has one or more *jump discontinuities*, or sudden changes in gradient, it can be shown that the Fourier series will normally converge to the average value of the two limiting values at the discontinuity. Even infinite discontinuities may be manageable.

As with infinite polynomial series, there are circumstances when the Fourier series expansion might not converge, but for most "realistic" signals or functions (details below...) the infinite Fourier series expansion converges to the function $x(t)$, or to the average value at a discontinuity.

The specific criteria for a Fourier series expansion to be valid are:

- $x(t)$ must be a single valued function.
- $x(t)$ must have only a finite number of maxima and minima.
- $x(t)$ must have only a finite number of discontinuities.
- $x(t)$ must be absolutely integrable over one period - that is

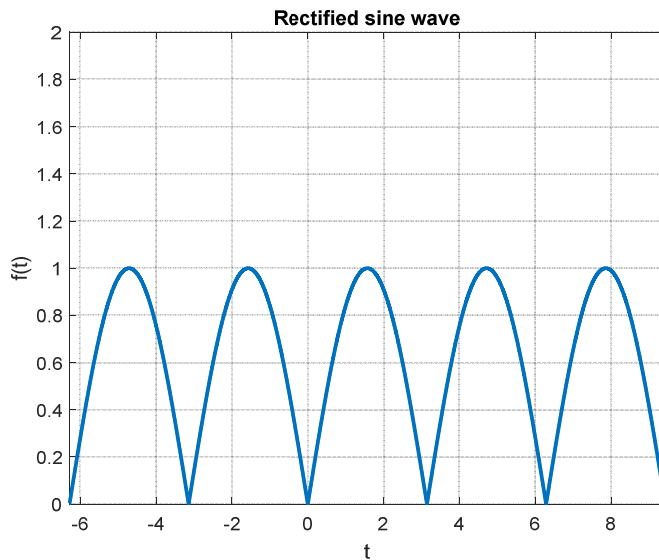
$$\int_0^T |x(t)| dt < \infty$$

In short:

- Fourier series are the appropriate tool for the *frequency analysis* of a periodic, continuous-time function - by expressing it as an infinite sum of sinusoidal functions.
- The signal/function is assumed to be infinitely long and perfectly periodic.
- The period of the signal/function defines the particular set of frequencies to be used for the Fourier series expansion.
- The function $x(t)$ can have multiple discontinuities in its value, gradient, or higher derivatives.
- The coefficients of the infinite series expansion can be obtained by integrating the function over one period after multiplying (modulating) it by any chosen basis function.

Example 1 : Sine-cosine Fourier series of a full-wave rectified sine wave.

Consider the infinite full-wave rectified sine wave function,
 $x(t) = |\sin(t)|$, illustrated below.



The signal $x(t)$ is periodic, with period $T = \pi$, with only one discontinuity in gradient per period and has a finite integral over a single period. So a Fourier series expansion can be applied to any one period - say the time interval $[0, \pi]$. In this case $\omega_0 = 2 \text{ rad s}^{-1}$ and the specific set of frequencies, for the Fourier series expansion are $n\omega_0 = 2n \text{ rad s}^{-1}$, where $n = 1, 2, 3, 4 \dots$

Hence
$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(2nt) + b_n \sin(2nt))$$

and the specific Fourier coefficients are given by

$$a_0 = \frac{1}{\pi} \int_0^{\pi} \sin(t) dt = \frac{1}{\pi} [-\cos(t)]_0^{\pi} = -\frac{1}{\pi} (-1 - 1) = \frac{2}{\pi}$$

$$\begin{aligned}
 a_n &= \frac{2}{\pi} \int_0^\pi \sin(t) \cos(2nt) dt \\
 &= \frac{1}{\pi} \int_0^\pi \left(\sin((1+2n)t) + \sin((1-2n)t) \right) dt \\
 &= \frac{1}{\pi} \left[-\frac{1}{1+2n} \cos((1+2n)t) - \frac{1}{1-2n} \cos((1-2n)t) \right]_0^\pi \\
 &= \frac{1}{\pi} \left(\left(\frac{1}{1+2n} + \frac{1}{1-2n} \right) - \left(-\frac{1}{1+2n} - \frac{1}{1-2n} \right) \right) \\
 &= \frac{1}{\pi} \frac{4}{1-4n^2}
 \end{aligned}$$

using the trigonometric identity:

$$\sin(\alpha) \cos(\beta) \equiv \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$$

Also

$$\begin{aligned}
 b_n &= \frac{2}{\pi} \int_0^\pi \sin(t) \sin(2nt) dt \\
 &= \frac{1}{\pi} \int_0^\pi \left(-\cos((1+2n)t) + \cos((1-2n)t) \right) dt \\
 &= \frac{1}{\pi} \left[-\frac{1}{1+2n} \sin((1+2n)t) + \frac{1}{1-2n} \sin((1-2n)t) \right]_0^\pi \\
 &= \frac{1}{\pi} ((0-0) - (0-0)) = 0
 \end{aligned}$$

using the trigonometric identity:

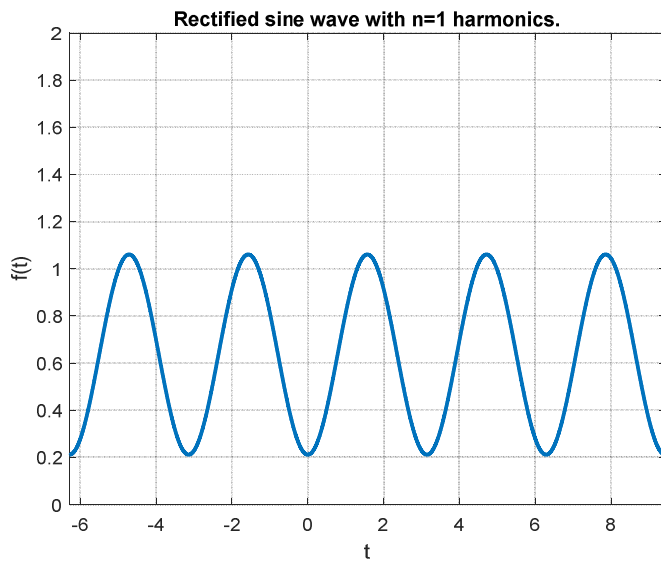
$$\sin(\alpha) \sin(\beta) \equiv \frac{1}{2} [-\cos(\alpha + \beta) + \cos(\alpha - \beta)]$$

Hence this particular function can be described as a constant plus pure cosine waves with frequencies that are multiples of $\omega_0 = 2 \text{ rad s}^{-1}$.

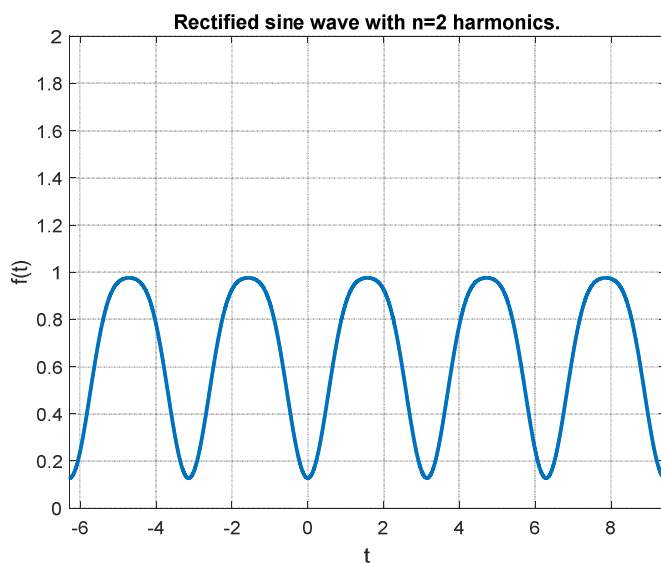
$$x(t) = \frac{2}{\pi} + \sum_{n=1}^{n=\infty} \frac{1}{\pi} \frac{4}{(1-4n^2)} \cos(2nt)$$

Note that it is entirely reasonable that this function should only be represented by cosine functions, since it is even over the period of integration. In general, the symmetry and structure of any function will be reflected in the nature of its Fourier series.

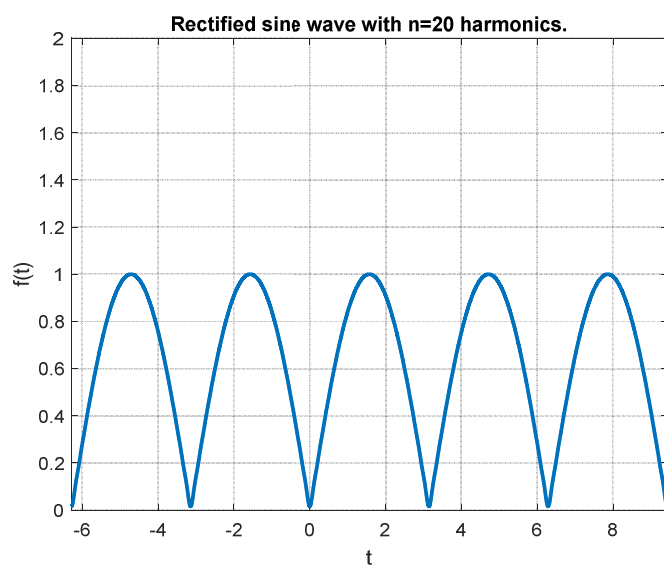
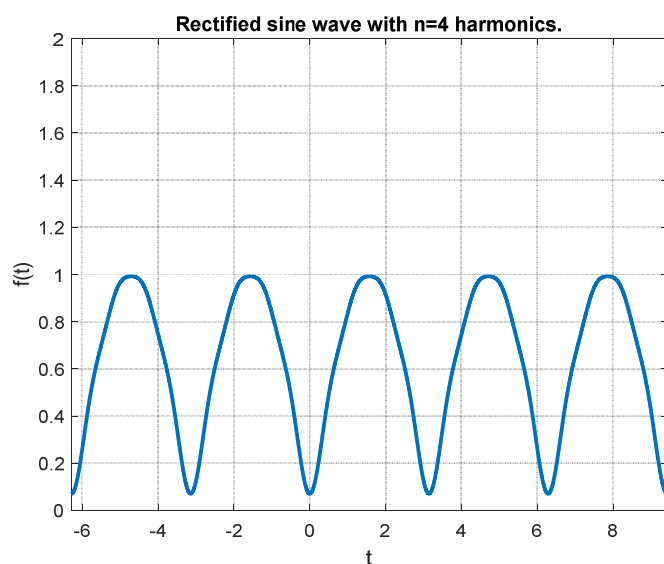
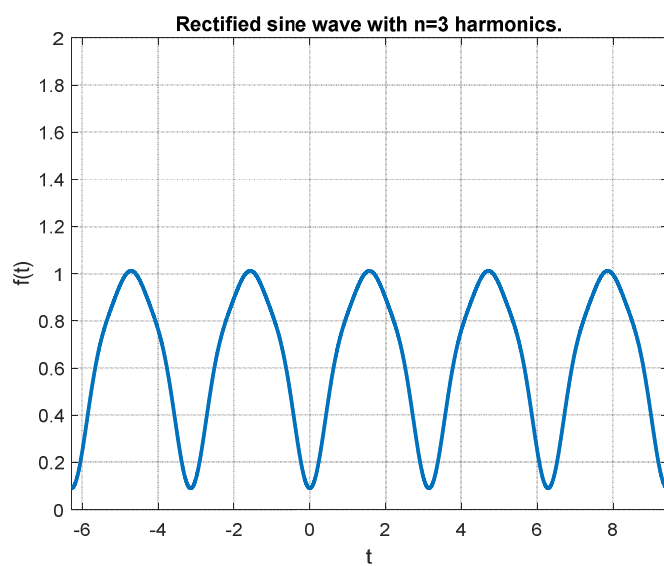
The Fourier expansion is exact for an infinite number of terms, but even just a few terms can represent a good approximation for simple functions such as this:



Above, just a constant and a single cosine - $\cos(2t)$ - is ineffective.



But adding just one more term - $\cos(2t)$ - is far better. Note that the biggest errors occur near sharp changes in gradient.

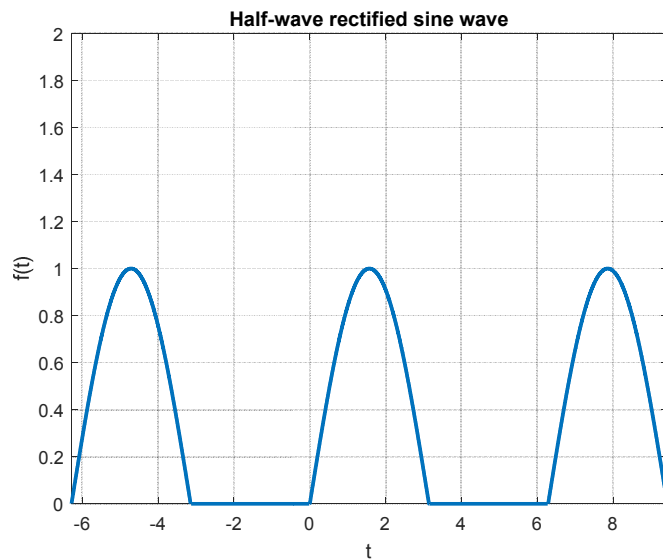


The approximation improves steadily as the appropriate amounts of higher-frequency content are added.

Example 2 : Sine-cosine Fourier series of a half-wave rectified sine wave.

Consider the infinite half-wave rectified sine wave function.

$$\begin{aligned} x(t) &= \sin(t) & 0 \leq t \leq \pi \\ x(t) &= 0 & \pi \leq t \leq 2\pi \end{aligned}$$



The signal $x(t)$ is periodic, with period $T = 2\pi$, and so a Fourier series expansion can be applied to any one period - say the time interval $[0, 2\pi]$. In this case $\omega_0 = 1 \text{ rad s}^{-1}$ and the specific set of frequencies, for the Fourier series expansion are $n\omega_0 = n \text{ rad s}^{-1}$, where $n = 1, 2, 3, 4 \dots$

Hence
$$x(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(nt) + b_n \sin(nt))$$

Since the function is zero over the interval $[\pi, 2\pi]$ the specific Fourier coefficients are given by

$$a_0 = \frac{1}{2\pi} \int_0^{\pi} \sin(t) dt = \frac{1}{2\pi} [-\cos(t)]_0^{\pi} = -\frac{1}{2\pi} (-1 - 1) = \frac{1}{\pi}$$

If $n = 1$

$$\begin{aligned}
 a_1 &= \frac{1}{2\pi} \int_0^\pi \sin(t) \cos(t) dt \\
 &= \frac{1}{4\pi} \int_0^\pi \sin(2t) dt = \frac{1}{4\pi} \left[-\frac{1}{2} \cos(2t) \right]_0^\pi = -\frac{1}{8\pi} (1-1) = 0
 \end{aligned}$$

and if $n > 1$

$$\begin{aligned}
 a_n &= \frac{2}{2\pi} \int_0^\pi \sin(t) \cos(nt) dt \\
 &= \frac{1}{2\pi} \int_0^\pi \left(\sin((1+n)t) + \sin((1-n)t) \right) dt \\
 &= \frac{1}{2\pi} \left[-\frac{1}{1+n} \cos((1+n)t) - \frac{1}{1-n} \cos((1-n)t) \right]_0^\pi \\
 &= \begin{cases} \frac{1}{2\pi} \left(\left(-\frac{1}{1+n} - \frac{1}{1-n} \right) - \left(-\frac{1}{1+n} - \frac{1}{1-n} \right) \right) = 0 & \text{when } n \text{ is odd} \\ \frac{1}{2\pi} \left(\left(+\frac{1}{1+n} + \frac{1}{1-n} \right) - \left(-\frac{1}{1+n} - \frac{1}{1-n} \right) \right) = \frac{2}{\pi} \frac{1}{(1-n^2)} & \text{when } n \text{ is even} \end{cases}
 \end{aligned}$$

using the trigonometric identity:

$$\sin(\alpha) \cos(\beta) \equiv \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$$

Also, provided that $n > 1$,

$$\begin{aligned}
 b_n &= \frac{2}{2\pi} \int_0^\pi \sin(t) \sin(nt) dt \\
 &= \frac{1}{2\pi} \int_0^\pi \left(-\cos((1+n)t) + \cos((1-n)t) \right) dt \\
 &= \frac{1}{2\pi} \left[-\frac{1}{1+n} \sin((1+n)t) + \frac{1}{1-n} \sin((1-n)t) \right]_0^\pi \\
 &= \frac{1}{2\pi} ((0-0) - (0-0)) = 0
 \end{aligned}$$

using the trigonometric identity:

$$\sin(\alpha) \sin(\beta) \equiv \frac{1}{2} [-\cos(\alpha + \beta) + \cos(\alpha - \beta)]$$

In the special case when $n = 1$

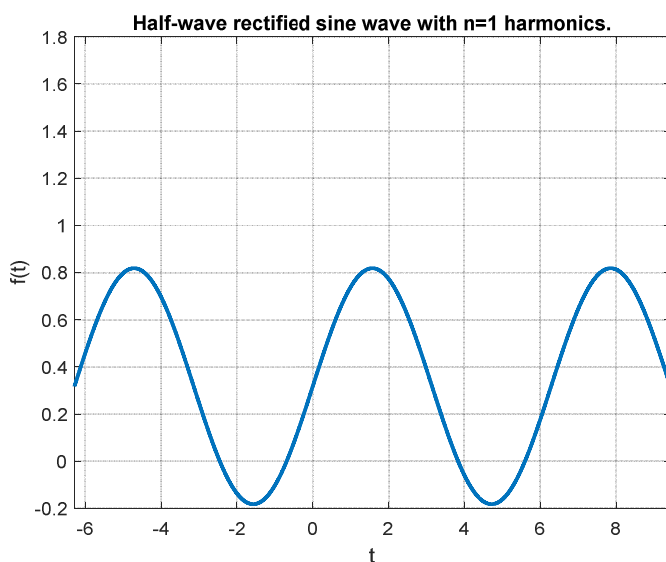
$$\begin{aligned}
 b_1 &= \frac{2}{2\pi} \int_0^\pi \sin(t) \sin(t) dt = \frac{1}{2\pi} \int_0^\pi (1 - \cos(2t)) dt \\
 &= \frac{1}{2\pi} [t - \sin(2t)]_0^\pi = \frac{1}{2\pi} ((\pi - 0) - (0 - 0)) = \frac{1}{2}
 \end{aligned}$$

Hence this particular function can be described as a Fourier series

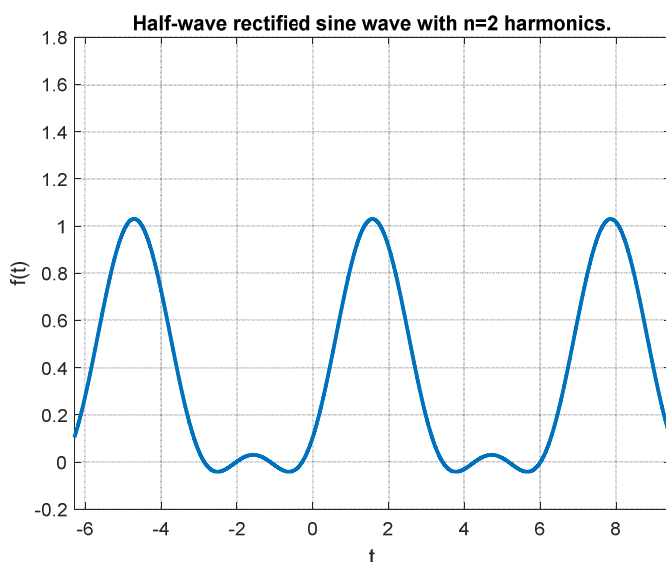
with frequencies that are multiples of $\omega_0 = 1 \text{ rad s}^{-1}$.

$$x(t) = \frac{1}{\pi} + \frac{1}{2} \sin(t) + \sum_{n=2,4,6,8}^{\infty} \frac{2}{\pi} \frac{1}{(1-n^2)} \cos(nt)$$

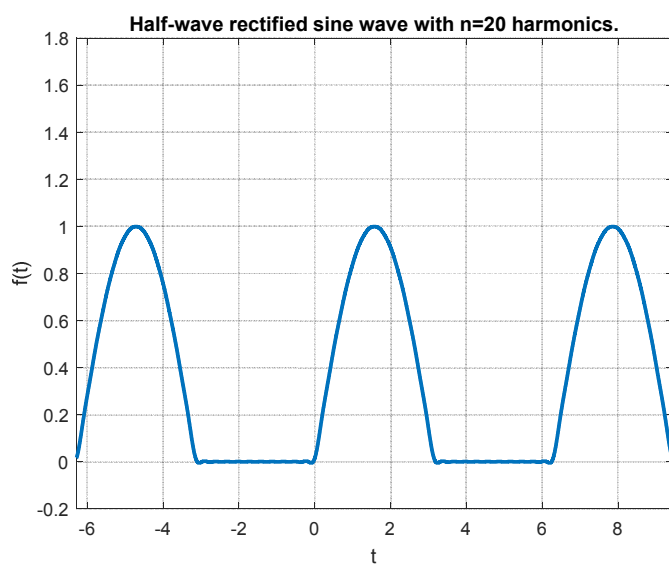
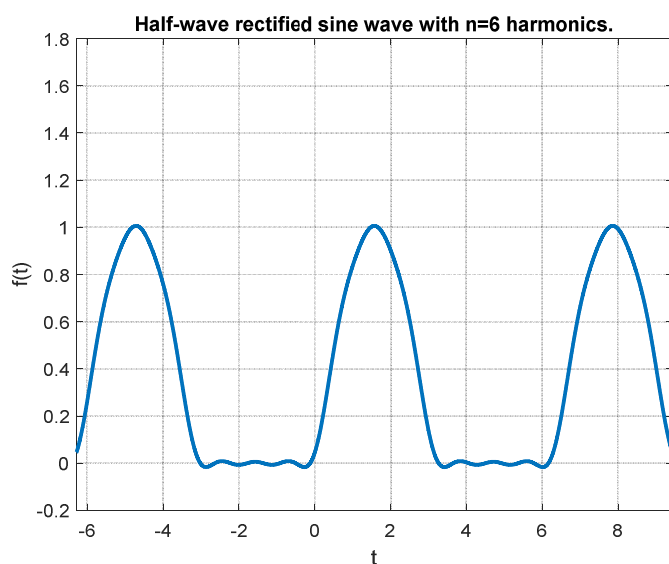
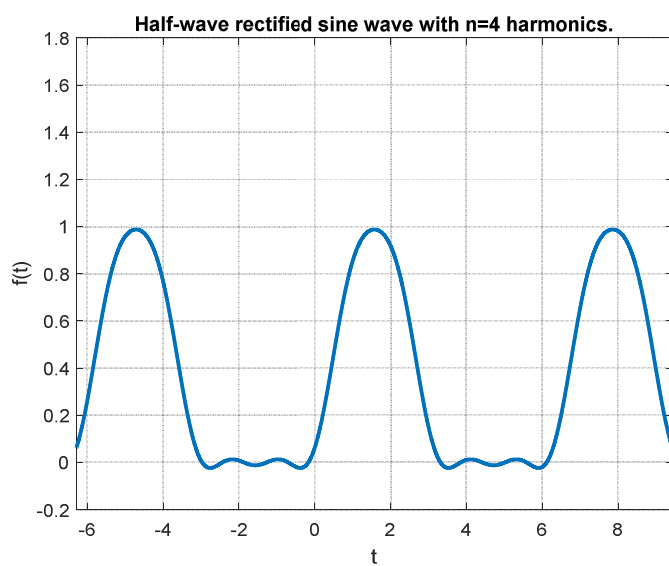
With just the first two terms, the graph looks like:



But with just one more term the shape improves radically:



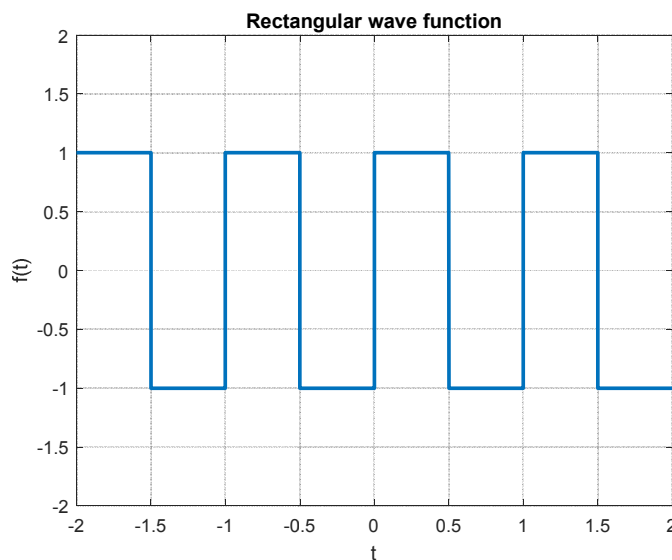
Beyond this point, there are no more contributions from odd sinusoidal terms, but the graphs improve steadily with four components, and then with six, etc.



Despite the improvement as the appropriate amounts of higher-frequency content are added, note the slight ripple that still remains.

Example 3 : Sine-cosine Fourier series of a rectangular wave

Consider the infinite pulse train (rectangular wave) function illustrated below.



The signal $x(t)$ is periodic, with period $T = 1$, and so a Fourier series expansion can be applied to any one period - say the time interval $[0, 1]$.

In this case $\omega_0 = 2\pi \text{ rad s}^{-1}$ and the specific set of frequencies, for the Fourier series expansion are $n\omega_0 = 2\pi n \text{ rad s}^{-1}$, where $n = 1, 2, 3, 4 \dots$

Hence
$$x(t) = a_0 + \sum_{n=1}^{n=\infty} (a_n \cos(2\pi nt) + b_n \sin(2\pi nt))$$

and the specific Fourier coefficients are given by

$$\begin{aligned} a_0 &= \frac{1}{1} \int_0^1 x(t) dt = \int_0^{0.5} 1 dt + \int_{0.5}^1 (-1) dt = 0 \\ a_n &= 2 \int_0^1 x(t) \cos(2\pi nt) dt \\ &= 2 \int_0^{0.5} \cos(2\pi nt) dt + 2 \int_{0.5}^1 (-1) \cos(2\pi nt) dt \\ &= 2 \left[\frac{1}{2\pi n} \sin(2\pi nt) \right]_0^{0.5} - 2 \left[\frac{1}{2\pi n} \sin(2\pi nt) \right]_{0.5}^1 \\ &= (0 - 0) - (0 - 0) = 0 \end{aligned}$$

and

$$\begin{aligned}
 b_n &= 2 \int_0^1 x(t) \sin(2\pi n t) dt \\
 &= 2 \int_0^{0.5} \sin(2\pi n t) dt + 2 \int_{0.5}^1 (-1) \sin(2\pi n t) dt \\
 &= 2 \left[-\frac{1}{2\pi n} \cos(2\pi n t) \right]_0^{0.5} - 2 \left[-\frac{1}{2\pi n} \cos(2\pi n t) \right]_{0.5}^1 \\
 &= -\frac{1}{\pi n} \left((\cos(\pi n) - 1) - (1 - \cos(\pi n)) \right) = \frac{2}{\pi n} (1 - \cos(\pi n))
 \end{aligned}$$

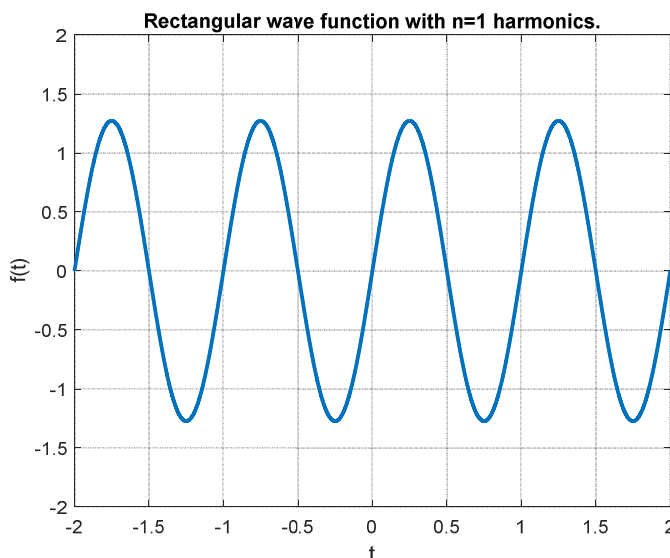
When n is even $\cos(\pi n) = 1$ and so $b_n = 0$.

When n is odd $\cos(\pi n) = -1$ and so $b_n = \frac{4}{\pi n}$.

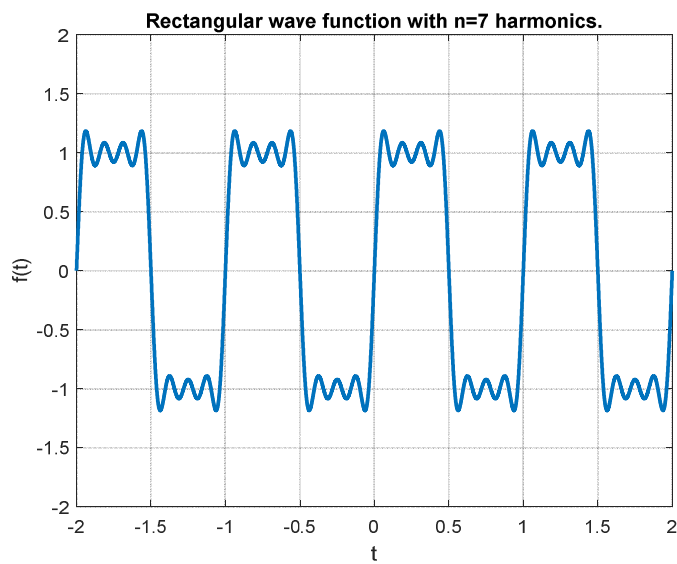
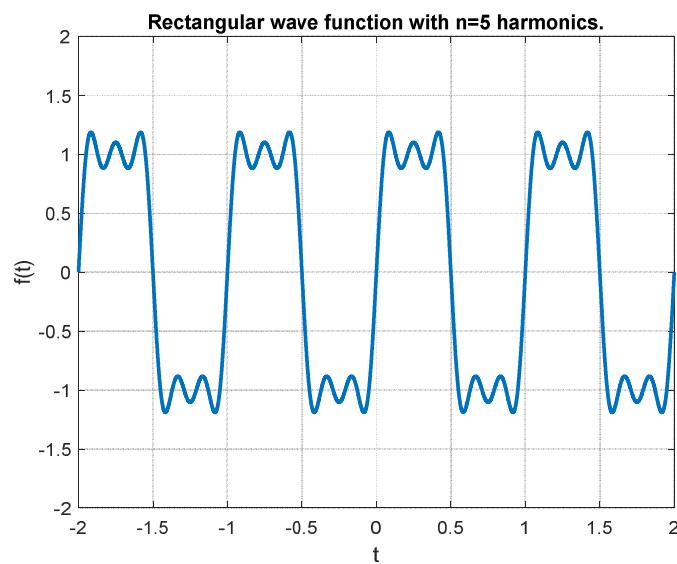
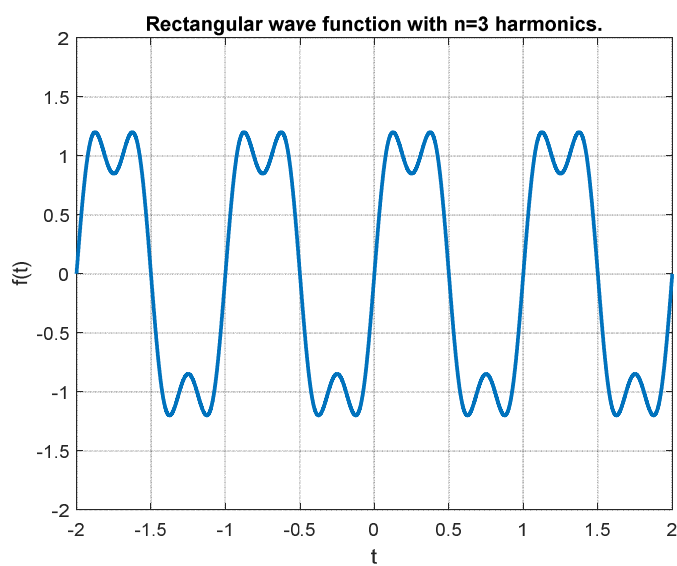
Hence this particular function can be described using only pure sine waves of odd multiple of the frequency $\omega_0 = 2\pi$.

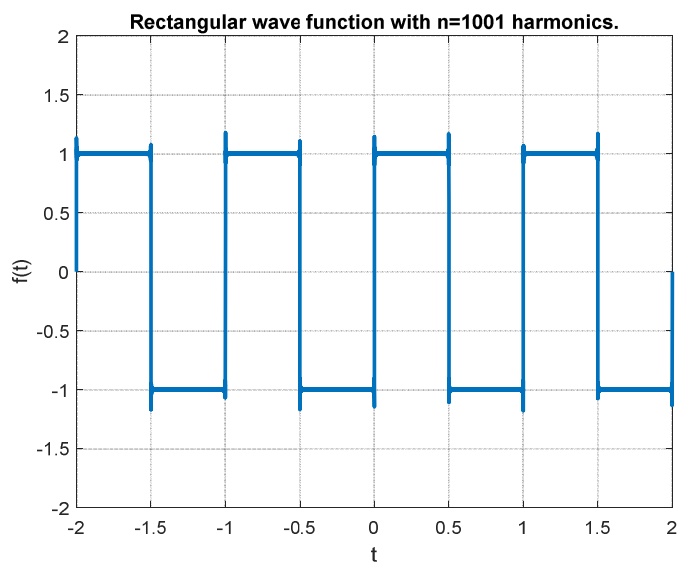
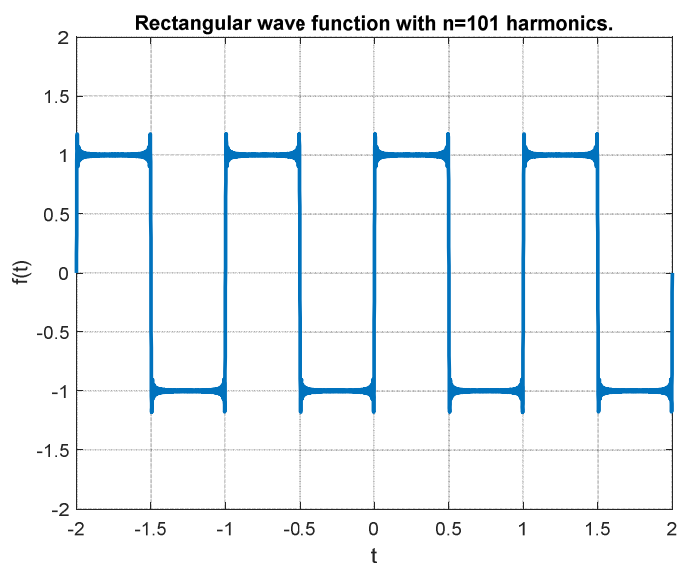
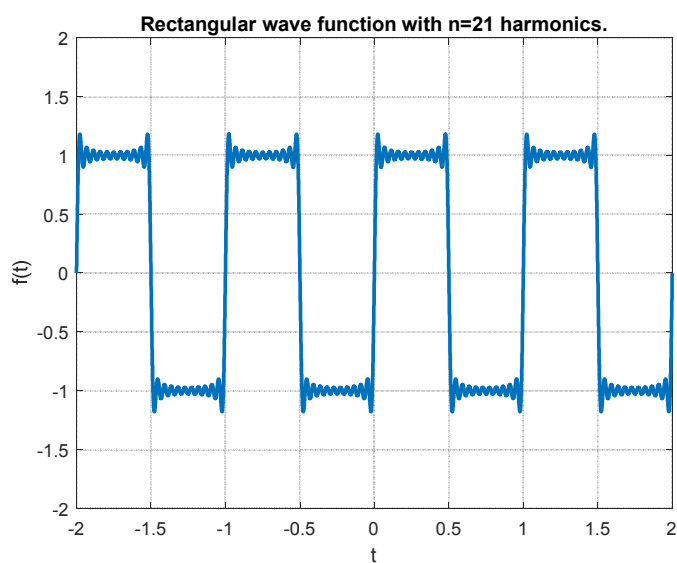
$$x(t) = \sum_{n=1,3,5,7,\dots}^{\infty} \frac{4}{\pi n} \sin(2\pi n t)$$

Again, since the function is odd over the chosen period of integration, it is not surprising that the series involves only sine waves.



Again, just using a single harmonic gives only a very crude representation of the pulse train signal, although the peaks and troughs are all in the correct positions. Adding further Fourier components helps to improve the shape steadily.





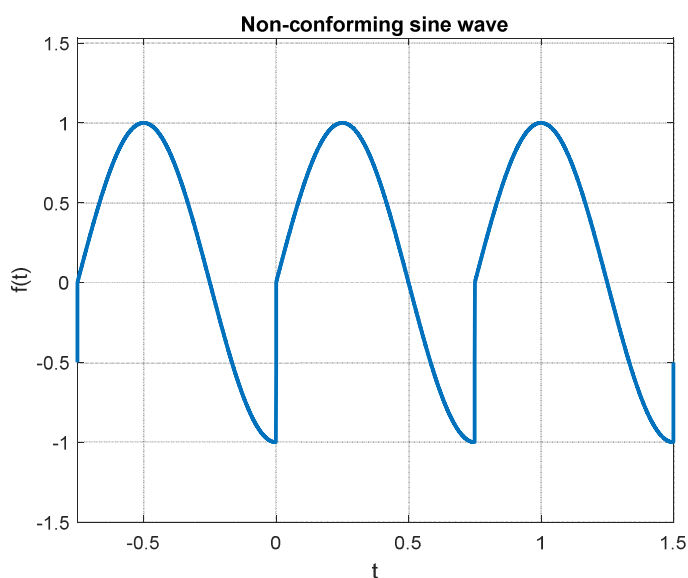
In this case, it can be seen that although the overall shape is excellent, even with large numbers of components, the Fourier series expansion is still failing to represent the function near the step discontinuities.

This overshoot near sudden changes in gradient is a general effect and is called the *Gibbs phenomenon*. In general, sudden changes in value are not well represented by sinusoidal waveforms.

At the discontinuity itself, when $t = 0.5$ s, the Fourier series will converge to the mid-point value, which in this particular case is zero.

Example 4 : Sine-cosine Fourier series of a partial sine wave

Consider the signal which is composed of portions of the 1 Hz sine wave function, $x(t) = \sin(2\pi t)$, but repeated every 0.75s, as illustrated below.



The signal $x(t)$ is periodic, with period $T = 0.75$, with only one discontinuity (in both value and gradient) per period and has a finite integral over a single period. So a Fourier series expansion can be applied to any one period - say the time interval $[0, 0.75]$. In this case

$\omega_0 = \frac{8}{3} \pi \text{ rad s}^{-1}$ and the specific set of angular frequencies, for the

Fourier series expansion are $n\omega_0 = \frac{8n\pi}{3} \text{ rad s}^{-1}$, where

$$n = 1, 2, 3, 4 \dots$$

Again
$$x(t) = a_0 + \sum_{n=1}^{n=\infty} (a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t))$$

where the specific Fourier coefficients are given by

$$\begin{aligned} a_0 &= \frac{1}{.75} \int_0^{.75} \sin(2\pi t) dt \\ &= \frac{4}{3} \frac{1}{2\pi} [-\cos(2\pi t)]_0^{.75} = \frac{2}{3\pi} (-(0-1)) = \frac{2}{3\pi} \end{aligned}$$

corresponding to a positive constant term.

Also

$$\begin{aligned} a_n &= \frac{2}{.75} \int_0^{.75} \sin(2\pi t) \cos\left(\frac{8\pi}{3} nt\right) dt \\ &= \frac{8}{3} \int_0^{.75} \frac{1}{2} \left(\sin\left(\left(2 + \frac{8}{3}n\right)\pi t\right) + \sin\left(\left(2 - \frac{8}{3}n\right)\pi t\right) \right) dt \\ &= \frac{4}{3} \left[-\frac{1}{\left(2 + \frac{8}{3}n\right)\pi} \cos\left(\left(2 + \frac{8}{3}n\right)\pi t\right) - \frac{1}{\left(2 - \frac{8}{3}n\right)\pi} \cos\left(\left(2 - \frac{8}{3}n\right)\pi t\right) \right]_0^{.75} \\ &= \frac{4}{3\pi} \left[-(0-0) - \left(-\frac{1}{\left(2 + \frac{8}{3}n\right)} - \frac{1}{\left(2 - \frac{8}{3}n\right)} \right) \right] \\ &= \frac{2}{\pi} \left(\frac{1}{(3+4n)} + \frac{1}{(3-4n)} \right) = \frac{12}{\pi(9-16n^2)} \end{aligned}$$

using the trigonometric identity:

$$\sin(\alpha) \cos(\beta) \equiv \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$$

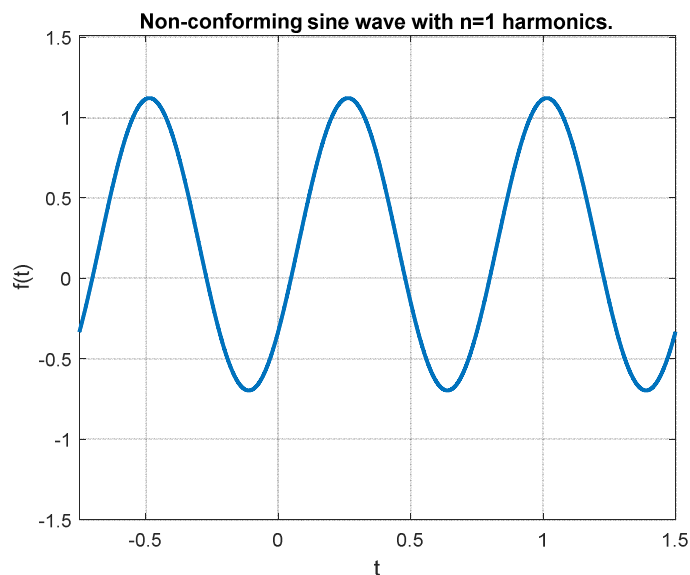
Similarly

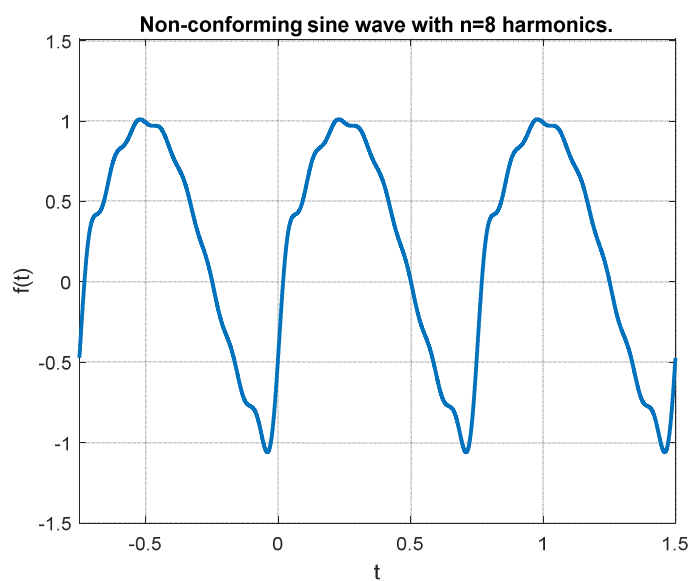
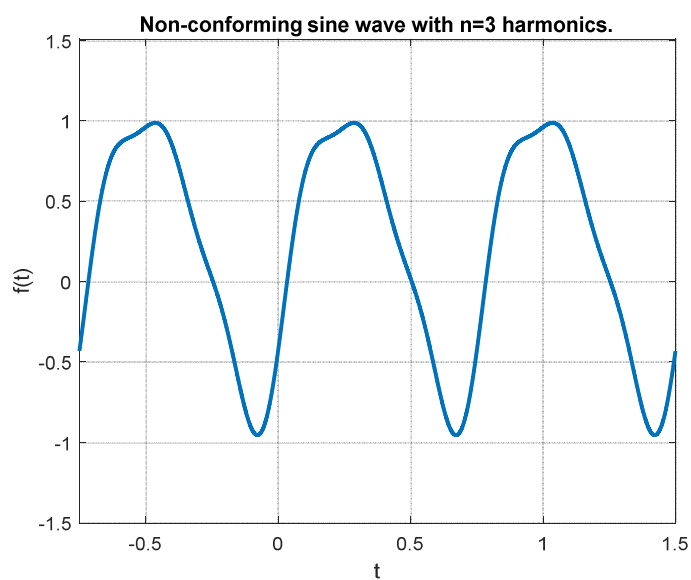
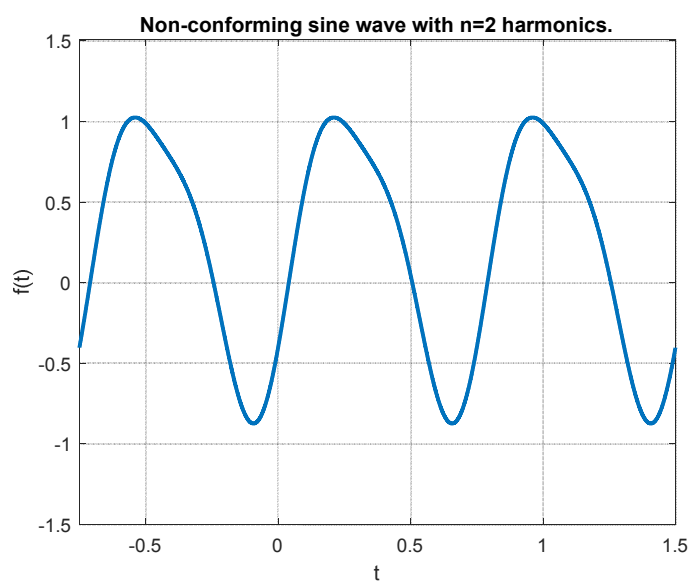
$$\begin{aligned}
 b_n &= \frac{2}{.75} \int_0^{.75} \sin(2\pi t) \sin\left(\frac{8\pi}{3} nt\right) dt \\
 &= \frac{8}{3} \int_0^{.75} \frac{1}{2} \left(-\cos\left(\left(2 + \frac{8}{3}n\right)\pi t\right) + \cos\left(\left(2 - \frac{8}{3}n\right)\pi t\right) \right) dt \\
 &= \frac{4}{3} \left[-\frac{1}{\left(2 + \frac{8}{3}n\right)\pi} \sin\left(\left(2 + \frac{8}{3}n\right)\pi t\right) + \frac{1}{\left(2 - \frac{8}{3}n\right)\pi} \sin\left(\left(2 - \frac{8}{3}n\right)\pi t\right) \right]_0^{.75} \\
 &= \frac{4}{3\pi} \left(\left(-\frac{1}{\left(2 + \frac{8}{3}n\right)} + \frac{1}{\left(2 - \frac{8}{3}n\right)} \right) - (0 - 0) \right) \\
 &= \frac{2}{\pi} \left(\frac{1}{(3 + 4n)} - \frac{1}{(3 - 4n)} \right) = -\frac{16n}{\pi(9 - 16n^2)}
 \end{aligned}$$

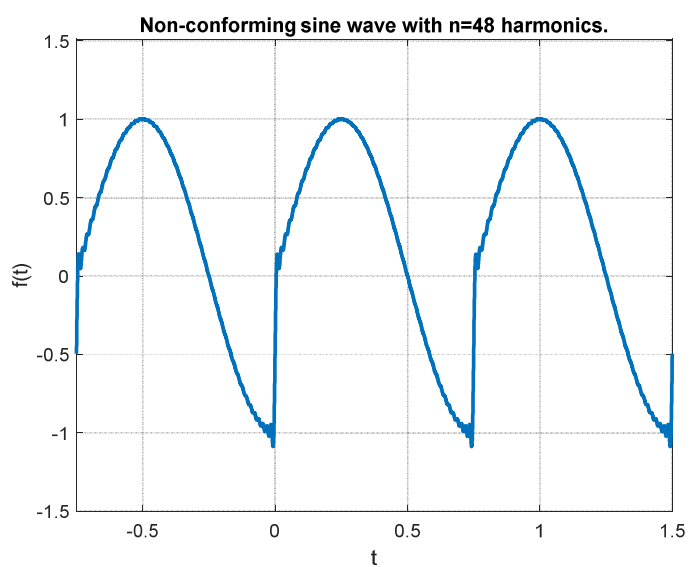
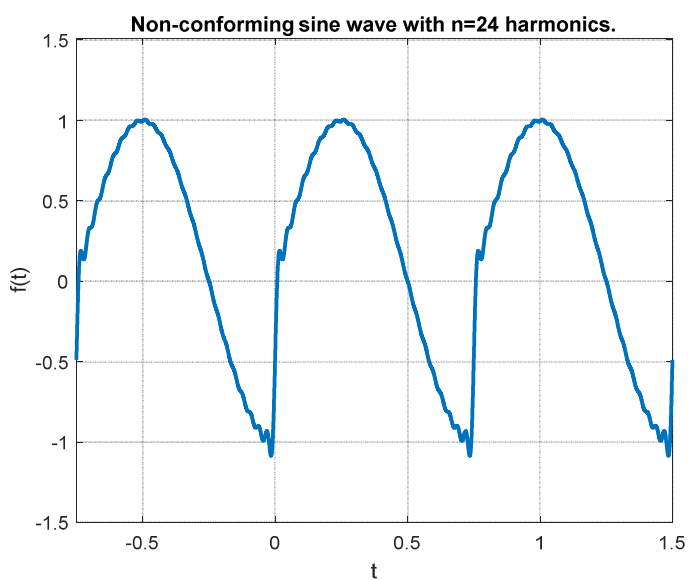
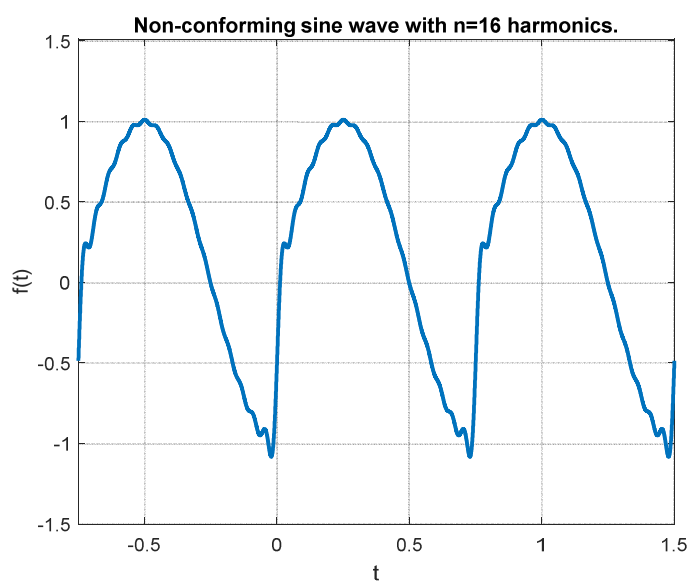
Hence this particular function can be described as a Fourier series expansion using multiples of the angular frequency $\omega_0 = \frac{8\pi}{3}$.

$$x(t) = \frac{2}{3\pi} + \sum_{n=1}^{n=\infty} \frac{12}{\pi(9 - 16n^2)} \cos\left(\frac{8n\pi}{3}t\right) - \sum_{n=1}^{n=\infty} \frac{16n}{\pi(9 - 16n^2)} \sin\left(\frac{8n\pi}{3}t\right)$$

Again, adding in frequency components steadily gives:







Many components are needed to cope with the edge discontinuity.

The impact of such edge effects is a major issue when framing a signal in order to use the short-time Fourier transform (STFT). In practice, since the frequency content of realistic signals will change over time, it is common to break up the signal into a series of consecutive frames, and then to apply a Fourier series expansion to each frame in sequence (deliberately and incorrectly!) assuming that each frame is part of a repeating, periodic sequence. This is called the STFT. But the choice of frame length is a user design decision. In the case above, despite the signal being a pure sine wave within each analysis frame, the periodicity of the frame did not happen to conform to that of the signal, and the consequent large edge discontinuity led to many additional frequency components appearing in the expansion that were not in the original signal. In effect, we were trying to build a 1 Hz signal out of many sinusoidal waves, and none of them were of 1 Hz frequency - the Fourier series frequencies are chosen by the length of the analysis frame, not by the signal content.

This problem is called *spectral leakage*, and in signal processing applications, is reduced by an additional process called *windowing*, which unfortunately introduces a different problem. In practice, engineering is often about compromises!

2.2. PURE SINE AND COSINE FOURIER SERIES

The sine-cosine Fourier series approach represents a periodic function as the sum of a constant and two independent infinite series - a cosine series and a sine series.

In general, if the origin is placed at the centre of the interval of one period of a periodic function, a function will be even over the interval if it satisfies the relationship

$$f(-t) = f(t) \quad (29)$$

for all values of the independent variable t .

Likewise, an odd function satisfies

$$f(-t) = -f(t) \quad (30)$$

over the interval.

In general, any function can be written as the sum of an even part and an odd part:

$$f(t) = \underbrace{\frac{1}{2}[f(t) + f(-t)]}_{\text{even component}} + \underbrace{\frac{1}{2}[f(t) - f(-t)]}_{\text{odd component}} \quad (31)$$

so that the constant and infinite series of cosine terms correspond to the even part, and the infinite series of sine terms correspond to the odd part.

It follows that if a function is even over the interval of integration, the coefficients of the sine terms will automatically be zero - the result is a pure *cosine Fourier series*.

Likewise a function which is odd over the interval of integration must result in a pure *sine Fourier series*.

2.3. AMPLITUDE-PHASE (HARMONICS) FORM OF FOURIER SERIES

In general, it is possible to write a mixture of a cosine and a sine signal as a single sinusoidal waveform of the same frequency, but scaled in amplitude and shifted in phase:

$$A \cos(\omega t) + B \sin(\omega t) \equiv C \sin(\omega t + \phi)$$

where

$$C = \sqrt{A^2 + B^2}, \quad \sin(\phi) = \frac{A}{C} \quad \text{and} \quad \cos(\phi) = \frac{B}{C}$$

For the general Fourier series expansion

$$x(t) = a_0 + \sum_{n=1}^{n=\infty} (a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t))$$

this result can be applied to each individual term in the expansion:

$$a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) = A_n \sin(n\omega_0 t + \phi_n)$$

where

$$A_n = \sqrt{a_n^2 + b_n^2}, \quad \sin(\phi) = \frac{a_n}{A_n} \quad \text{and} \quad \cos(\phi) = \frac{b_n}{A_n}$$

so that the Fourier series can be written in an alternative amplitude-phase form as:

$$x(t) = a_0 + \sum_{n=1}^{n=\infty} A_n \sin(n\omega_0 t + \phi_n)$$

In this form, $A_n = \sqrt{a_n^2 + b_n^2}$ quantifies the overall amplitude of each component of angular frequency $n\omega_0$, including contributions from both the sine and cosine parts.

This form is also called the *harmonics form* of a Fourier series, stressing the fact that the signal is represented as the linear sum of a weighted set of sinusoids with frequencies which are integer multiples of the fundamental angular frequency $\omega_0 = \frac{2\pi}{T}$. This parallels the idea from music that an ideal tuned string will vibrate in a fashion that can be described a mixture of a fundamental frequency f_0 and a series of harmonic partials (overtones) with frequencies that are integer multiples of the fundamental.

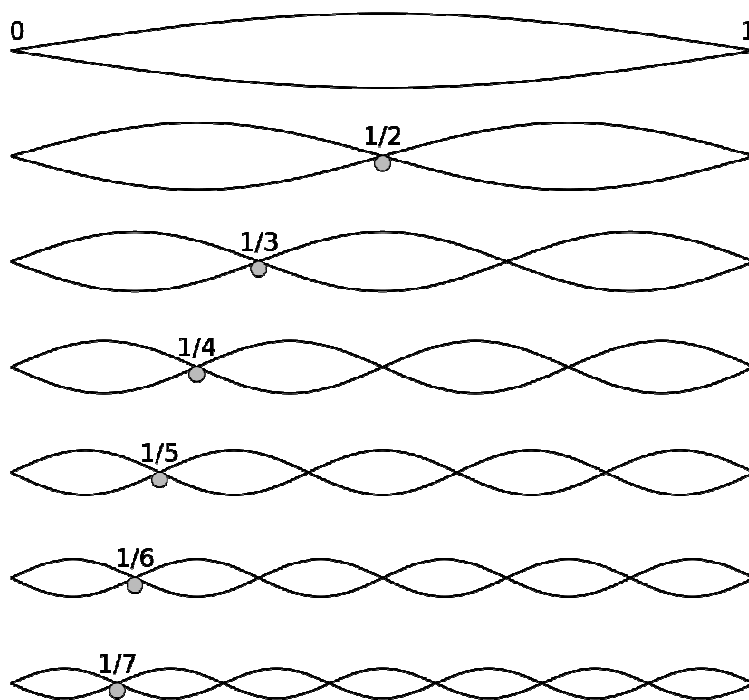


Figure 6 The first seven modes of vibration of a perfect stretched string, corresponding to the harmonic frequencies nf_0 where $n = 1, 2, 3, 4, 5, 6, 7$.

These shapes describe some of the *modes of vibration*, corresponding to the first seven harmonic partials of a perfect string (the fundamental is the first partial), but the general idea of *modal analysis* is also applied widely, to mechanical and electrical systems for example.

2.4. EXPONENTIAL FOURIER SERIES

The trigonometric functions, sine and cosine, can be defined algebraically in terms of complex exponentials as:

$$\sin(\theta) \triangleq \frac{e^{j\theta} - e^{-j\theta}}{2j} \quad (32)$$

$$\cos(\theta) \triangleq \frac{e^{j\theta} + e^{-j\theta}}{2} \quad (33)$$

Using these expressions in the general Fourier series expansion

$$x(t) = a_0 + \sum_{n=1}^{n=\infty} (a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t))$$

gives

$$\begin{aligned} x(t) &= a_0 + \sum_{n=1}^{n=\infty} \left(a_n \frac{e^{jn\omega_0 t} + e^{-jn\omega_0 t}}{2} + b_n \frac{e^{jn\omega_0 t} - e^{-jn\omega_0 t}}{2j} \right) \\ &= a_0 + \sum_{n=1}^{n=\infty} \frac{1}{2} \left((a_n - jb_n) e^{jn\omega_0 t} + (a_n + jb_n) e^{-jn\omega_0 t} \right) \end{aligned}$$

can be written in a much more compact form as

$$\begin{aligned} x(t) &= a_0 + \sum_{n=1}^{n=\infty} \left(a_n \frac{e^{jn\omega_0 t} + e^{-jn\omega_0 t}}{2} + b_n \frac{e^{jn\omega_0 t} - e^{-jn\omega_0 t}}{2j} \right) \\ &= a_0 + \sum_{n=1}^{n=\infty} \frac{1}{2} (a_n - jb_n) e^{jn\omega_0 t} + \sum_{n=1}^{n=\infty} \frac{1}{2} (a_n + jb_n) e^{-jn\omega_0 t} \\ &= a_0 + \sum_{n=1}^{n=\infty} \frac{1}{2} (a_n - jb_n) e^{jn\omega_0 t} + \sum_{n=-\infty}^{n=-1} \frac{1}{2} (a_n + jb_n) e^{jn\omega_0 t} \\ &= \sum_{n=-\infty}^{n=\infty} c_n e^{jn\omega_0 t} \end{aligned}$$

where

$$c_n = \frac{1}{2} (a_n - jb_n) \quad \text{for positive } n$$

and

$$c_{-n} = \frac{1}{2} (a_n + jb_n) = \overline{c_n}$$

Also

$$c_0 = a_0$$

Overall,

$$x(t) = \sum_{n=-\infty}^{n=\infty} c_n e^{jn\omega_0 t} \quad (34)$$

defines the *complex exponential form* of the Fourier series expansion where the complex coefficients c_n can be calculated in a single step

from:

$$\begin{aligned}
 c_n &= a_n - jb_n \\
 &= \frac{1}{2} \left(\frac{2}{T} \int_{\langle T \rangle} x(t) \cos(n\omega_0 t) dt - j \frac{2}{T} \int_{\langle T \rangle} x(t) \sin(n\omega_0 t) dt \right) \\
 &= \frac{1}{T} \int_{\langle T \rangle} x(t) (\cos(n\omega_0 t) - j \sin(n\omega_0 t)) dt \\
 &= \frac{1}{T} \int_{\langle T \rangle} x(t) \exp(-jn\omega_0 t) dt
 \end{aligned}$$

The expression

$$x(t) = \sum_{n=-\infty}^{n=\infty} c_n e^{jn\omega_0 t}.$$

is sometimes referred to as the *Fourier synthesis*, stage while evaluating the coefficients via

$$c_n = \frac{1}{T} \int_{\langle T \rangle} x(t) \exp(-jn\omega_0 t) dt$$

is the *Fourier analysis* stage.

As with the sine and cosine functions, the complex exponentials form an orthogonal basis set since

$$\langle e^{jm\omega_0 t}, e^{jn\omega_0 t} \rangle = \int_{-T/2}^{T/2} e^{-jm\omega_0 t} e^{jn\omega_0 t} dt = \int_{-T/2}^{T/2} e^{j(m-n)\omega_0 t} dt$$

If $m \neq n$

$$\begin{aligned}
 \int_{-T/2}^{T/2} e^{j(m-n)\omega_0 t} dt &= \left[\frac{1}{j(m-n)\omega_0} e^{j(m-n)\omega_0 t} \right]_{-T/2}^{T/2} \\
 &= \frac{1}{j(m-n)\omega_0} e^{j(m-n)\pi} - \frac{1}{j(m-n)\omega_0} e^{-j(m-n)\pi} = 0
 \end{aligned}$$

and if $m = n$

$$\int_{-T/2}^{T/2} e^{j(m-n)\omega_0 t} dt = \int_{-T/2}^{T/2} 1 dt = [t]_{-T/2}^{T/2} = \frac{T}{2} - -\frac{T}{2} = T$$

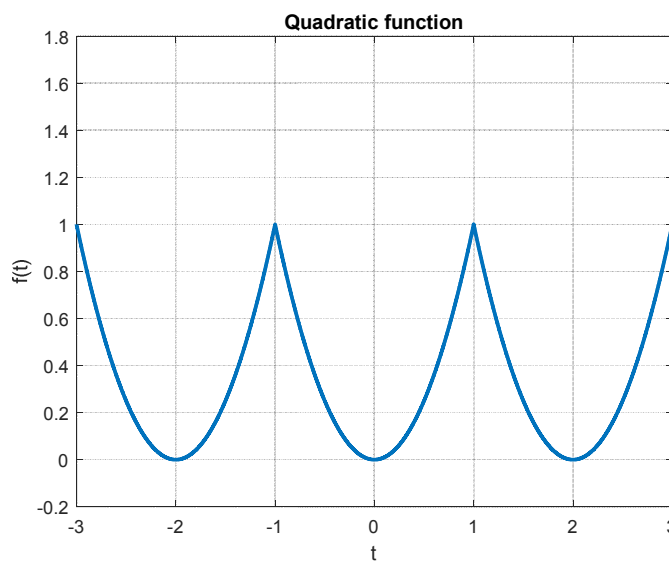
so that the complex exponentials form an orthogonal set of functions.

The various forms of Fourier series are equivalent - it is true to say that

$$\begin{aligned} x(t) &= a_0 + \sum_{n=1}^{n=\infty} \left(a_n \cos(2\pi nt) + b_n \sin(2\pi nt) \right) \\ &= a_0 + \sum_{n=1}^{n=\infty} A_n \sin(n\omega_0 t + \phi_n) \\ &= \sum_{n=-\infty}^{n=\infty} c_n e^{jn\omega_0 t} \end{aligned}$$

Example 5 : The exponential Fourier series of a parabola

Consider the Fourier series of the periodic function $x(t) = t^2$, defined on the interval $[-1, 1]$.



Overall, in exponential form,

$$x(t) = \sum_{n=-\infty}^{n=\infty} c_n e^{jn\omega_0 t},$$

where $\omega_0 = \frac{2\pi}{2} = \pi$, and $c_n = \frac{1}{2} \int_{-1}^1 t^2 \exp(-jn\pi t) dt$

For $n \neq 0$, integrating by parts gives

$$\begin{aligned}
c_n &= \frac{1}{2} \int_{-1}^1 t^2 \exp(-jn\pi t) dt \\
&= \frac{1}{2} \left[\frac{1}{-jn\pi} t^2 \exp(-jn\pi t) \right]_{-1}^1 - \frac{1}{2} \int_{-1}^1 2t \left(\frac{1}{-jn\pi} \exp(-jn\pi t) \right) dt \\
&= \frac{1}{2} \left[\frac{1}{-jn\pi} t^2 \exp(-jn\pi t) \right]_{-1}^1 + \frac{1}{jn\pi} \int_{-1}^1 t \left(\exp(-jn\pi t) \right) dt \\
&= \left[j \frac{1}{2n\pi} t^2 \exp(-jn\pi t) \right]_{-1}^1 + \frac{1}{jn\pi} \left[\frac{1}{-jn\pi} t \exp(-jn\pi t) \right]_{-1}^1 - \int_{-1}^1 \frac{1}{-jn\pi} \left(\exp(-jn\pi t) \right) dt \\
&= \left[-\frac{1}{2jn\pi} t^2 \exp(-jn\pi t) + \frac{1}{n^2\pi^2} t \exp(-jn\pi t) + \frac{1}{jn^3\pi^3} \exp(-jn\pi t) \right]_{-1}^1
\end{aligned}$$

and using the limits gives

$$\begin{aligned}
c_n &= \frac{1}{n\pi} \sin(n\pi) + \frac{2}{n^2\pi^2} \cos(n\pi) - \frac{2}{n^3\pi^3} \sin(n\pi) \\
&= \frac{2}{n^2\pi^2} \cos(n\pi) = \frac{2}{n^2\pi^2} (-1)^n
\end{aligned}$$

When $n = 0$

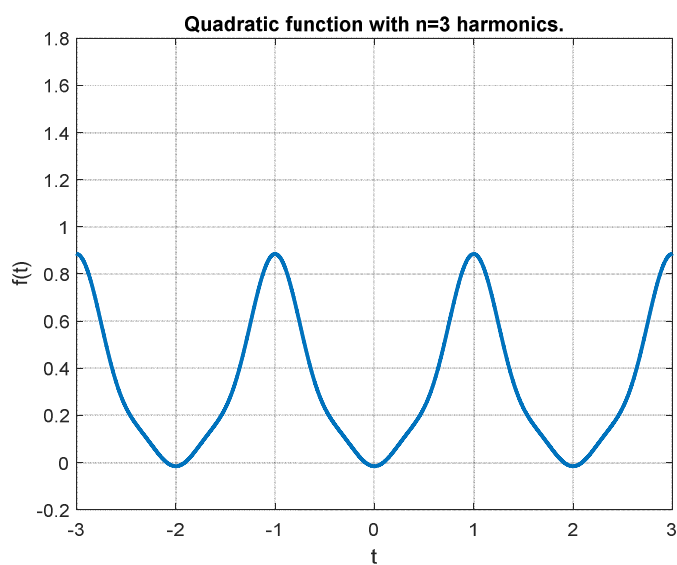
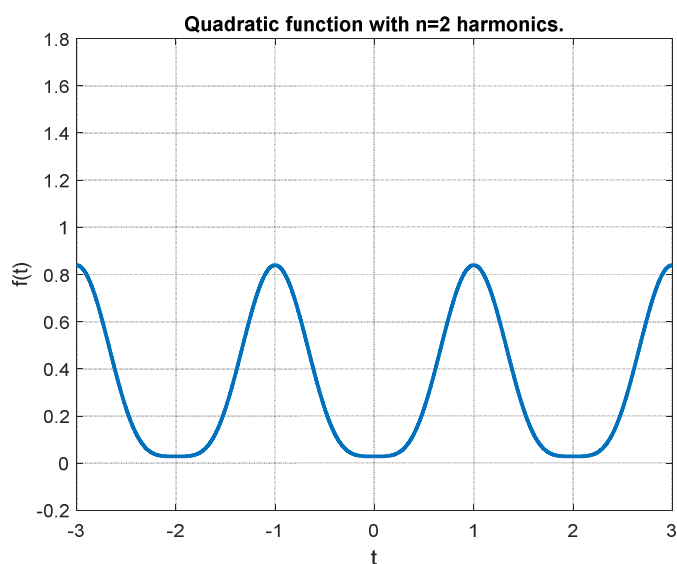
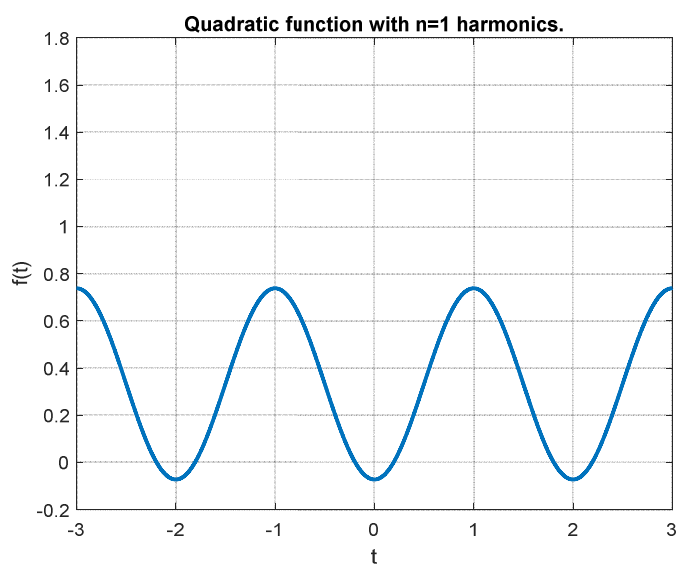
$$c_0 = \frac{1}{2} \int_{-1}^1 t^2 dt = \frac{1}{2} \left[\frac{t^3}{3} \right]_{-1}^1 = \frac{1}{2} \left(\frac{1}{3} - -\frac{1}{3} \right) = \frac{1}{3}$$

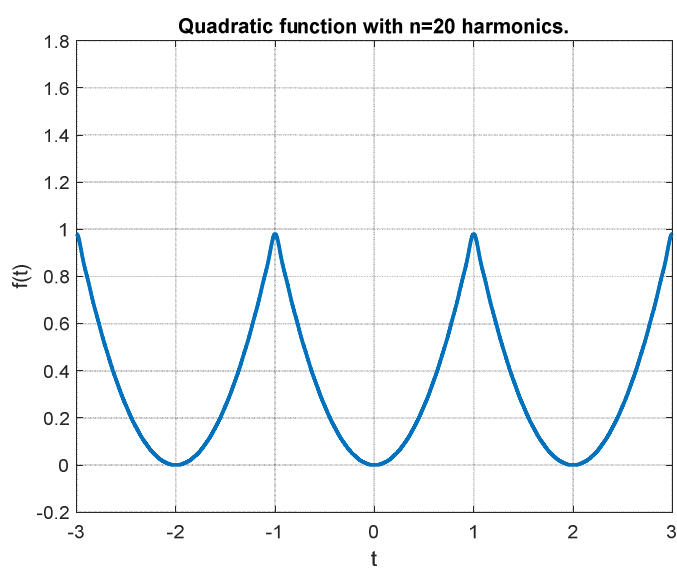
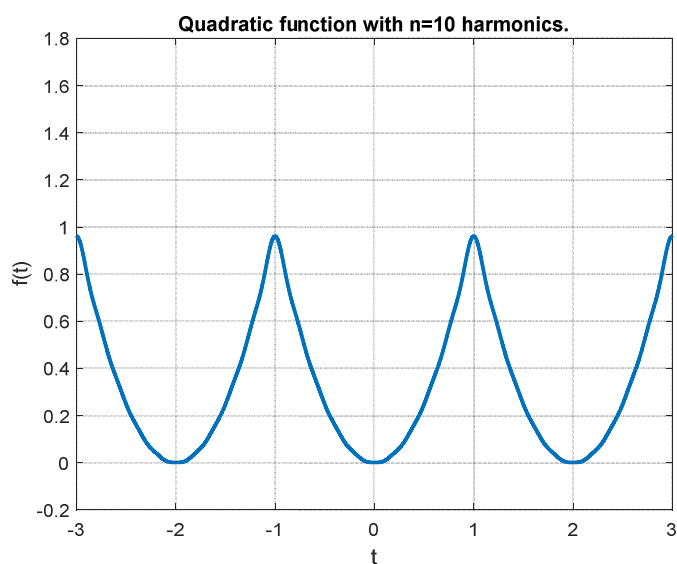
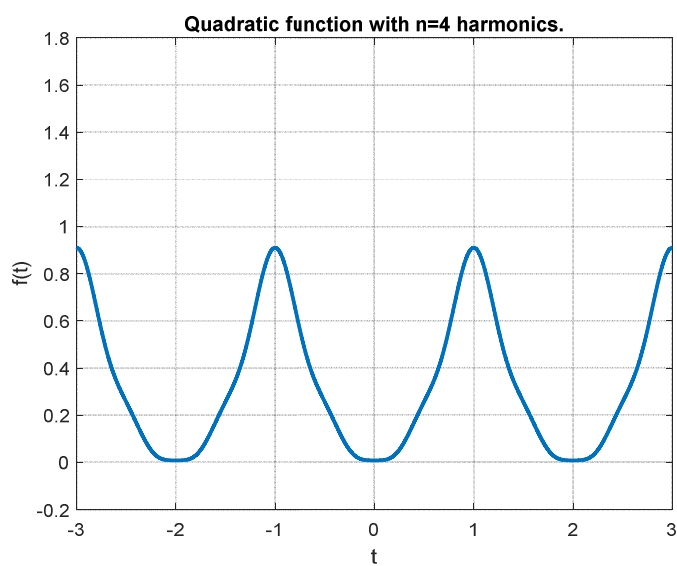
Hence, overall,

$$\begin{aligned}
x(t) &= \frac{1}{3} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi^2} (-1)^n e^{jn\pi t} + \sum_{n=1}^{\infty} \frac{2}{(-n)^2\pi^2} (-1)^{-n} e^{-jn\pi t} \\
&= \frac{1}{3} + \sum_{n=1}^{\infty} \frac{2}{n^2\pi^2} (-1)^n (e^{jn\pi t} + e^{-jn\pi t}) \\
&= \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} (-1)^n \cos(n\pi t)
\end{aligned}$$

The function is even with respect to the interval and so we would expect the result to involve pure cosine terms.

Adding components steadily produces:





2.5. THE LINE SPECTRUM OF A PERIODIC SIGNAL

For a Fourier series expansion, the moduli of the coefficients,

$$|c_n| = \frac{1}{2} \sqrt{a_n^2 + b_n^2} \text{ in complex exponential form}$$

and $A_n = \sqrt{a_n^2 + b_n^2}$ in amplitude-phase form

provide a measure of the total amount of each sinusoid contained within the original signal at each specific frequency. In any case, the orthogonality of the basis functions ensures that these measures are independent for different frequencies.

It is common to plot the variations in the amplitudes A_n as a *line spectrum*, as illustrated below.

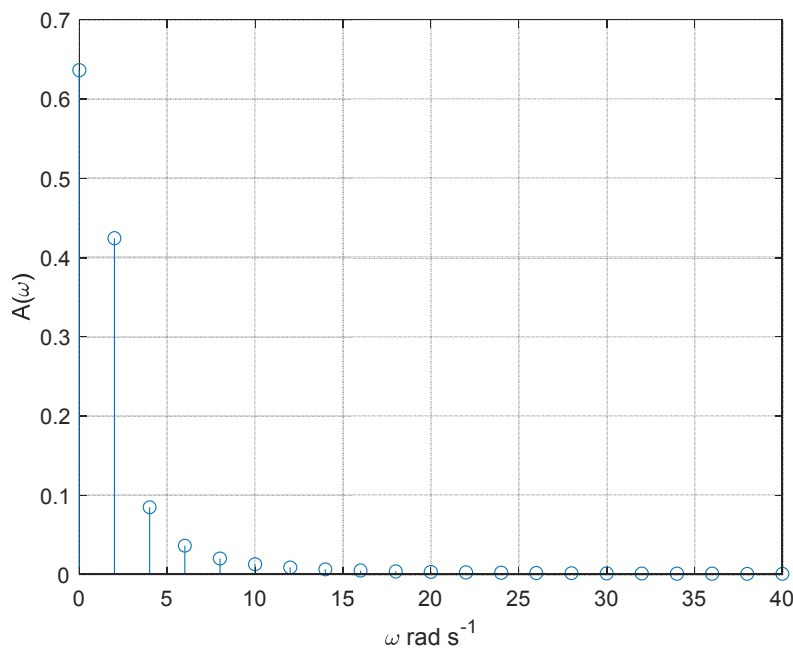


Figure 7 The line spectrum for Example 1.
The Fourier series of a full-wave rectified sine wave, where

$$x(t) = \frac{2}{\pi} + \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{(1 - 4n^2)} \cos(2nt)$$

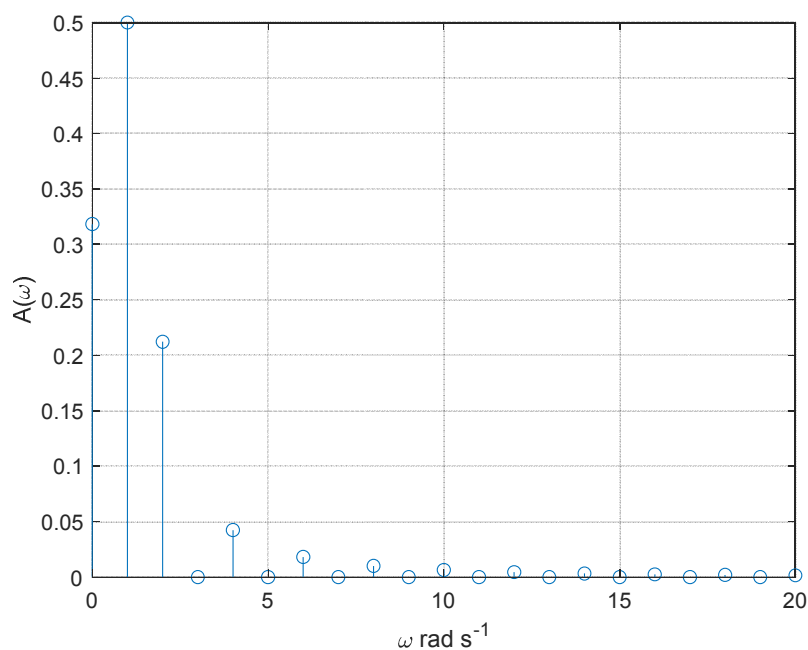


Figure 8 The line spectrum for Example 2.
The Fourier series of a half-wave rectified sine wave, where

$$x(t) = \frac{1}{\pi} + \frac{1}{2} \sin(t) + \frac{2}{\pi} \sum_{n=2,4,6,8}^{\infty} \frac{1}{(1-n^2)} \cos(nt)$$

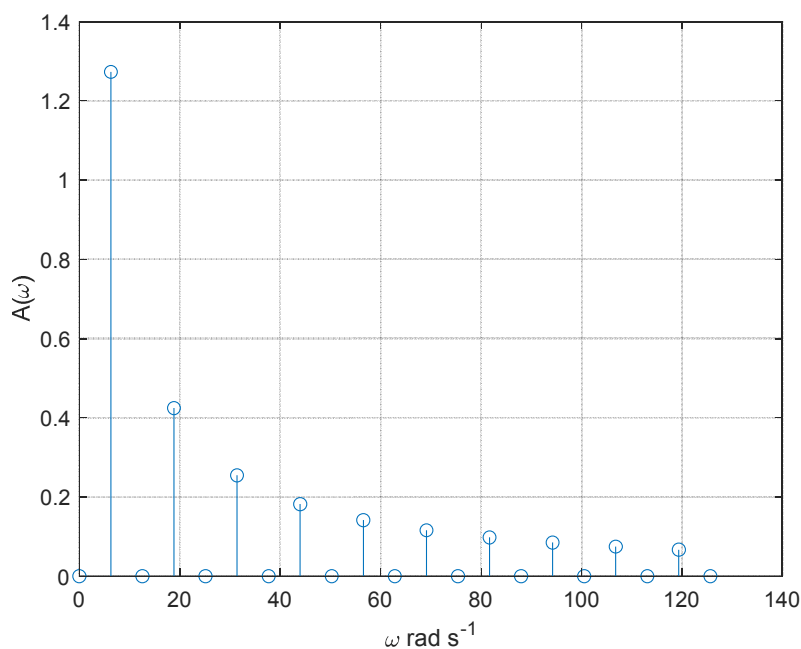


Figure 9 The line spectrum for Example 3.
The Fourier series of a rectangular wave function, where

$$x(t) = \sum_{n=1,3,5,7,\dots}^{\infty} \frac{4}{\pi n} \sin(2\pi n t)$$

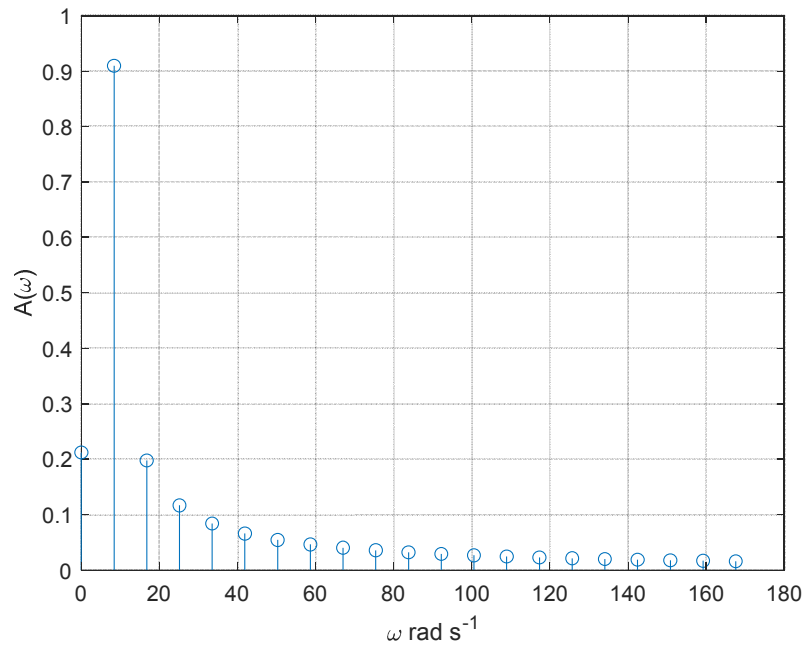


Figure 10 The line spectrum for Example 4.

The Fourier series of a partial sine wave, where

$$x(t) = \frac{2}{3\pi} + \sum_{n=1}^{n=\infty} \frac{12}{\pi(9-16n^2)} \cos\left(\frac{8\pi n}{3}t\right) - \sum_{n=1}^{n=\infty} \frac{16n}{\pi(9-16n^2)} \sin\left(\frac{8\pi n}{3}t\right)$$

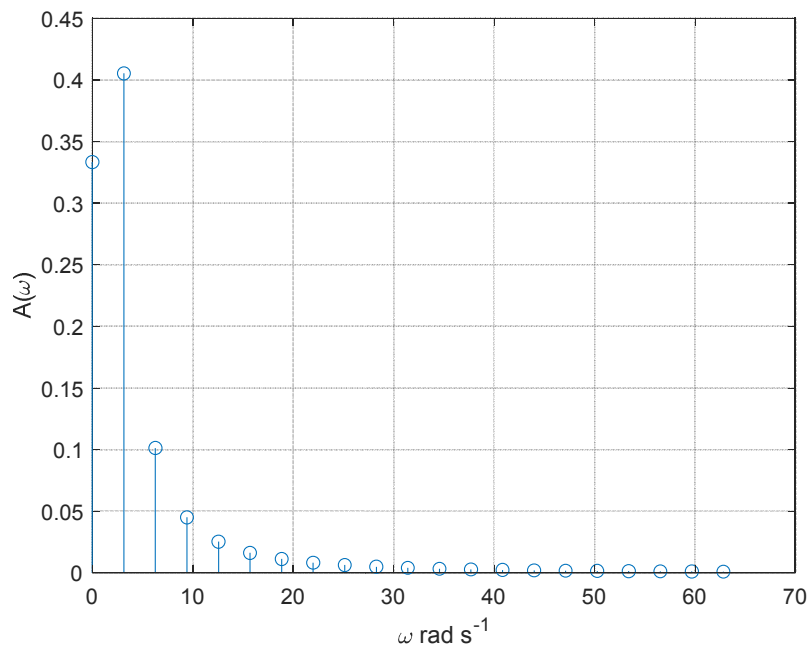


Figure 11 The line spectrum for Example 5. The Fourier series of

a parabola, where $x(t) = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=1}^{n=\infty} \frac{1}{n^2} (-1)^n \cos(n\pi t)$

2.6. SPECTRAL ENERGY AND PARSEVAL'S THEOREM

For a periodic function $x(t)$ with period T , the line spectrum describes the amplitudes of all of the chosen sinusoidal (and hence orthogonal) Fourier basis functions, $\sin(n\omega_0 t + \phi_n)$, where the fundamental frequency is $\omega_0 = \frac{2\pi}{T}$.

This discrete spectrum arises because of the periodic nature of the original function - the basis functions have to fit the repeat interval precisely. But another consequence of the perfect periodicity of $x(t)$ is that the signal has infinite energy - the signal energy is only finite during a single period. A signal of this form is called a *power signal*.

In general, the *instantaneous power* for a signal is:

$$P(t) = |x(t)|^2 \quad (35)$$

and the *total energy* over a single period of a periodic signal is

$$E = \int_{\langle T \rangle} |x(t)|^2 dt \quad (36)$$

so that the average power over a single period is

$$P = \frac{1}{T} \int_{\langle T \rangle} |x(t)|^2 dt \quad (37)$$

Using the general Fourier exponential series expansion

$$x(t) = \sum_{n=-\infty}^{n=\infty} c_n e^{jn\omega_0 t}$$

means that we can write

$$\begin{aligned} E &= \int_{\langle T \rangle} |x(t)|^2 dt \\ &= \int_{\langle T \rangle} \left(\sum_{n=-\infty}^{n=\infty} c_n e^{jn\omega_0 t} \right) \left(\sum_{m=-\infty}^{m=\infty} \overline{c_m} e^{-jm\omega_0 t} \right) dt \\ &= \int_{\langle T \rangle} \sum_{n=-\infty}^{n=\infty} \sum_{m=-\infty}^{m=\infty} c_n \overline{c_m} e^{jn\omega_0 t} e^{-jm\omega_0 t} dt \\ &= \sum_{n=-\infty}^{n=\infty} \sum_{m=-\infty}^{m=\infty} c_n \overline{c_m} \int_{\langle T \rangle} e^{-jm\omega_0 t} e^{jn\omega_0 t} dt = \sum_{n=-\infty}^{n=\infty} |c_n|^2 T = T \sum_{n=-\infty}^{n=\infty} |c_n|^2 \end{aligned}$$

This result is called *Parseval's Theorem* - it represents the fact that the

average power of the signal over a single period is directly equivalent to the sum of the squares of the coefficients associated with each complex frequency component:

$$\frac{1}{T} \int_{\langle T \rangle} |x(t)|^2 dt = \sum_{n=-\infty}^{n=\infty} |c_n|^2 \quad (38)$$

Mathematically, this represents the fact that the Fourier series is a *unitary transformation* - there is no gain/scaling associated with the transformation between the time and frequency domains. More intuitively, the signal energy over a period is the same whether the signal is viewed in the time or frequency domains.

In terms of the trigonometric Fourier series expansion

$$x(t) = a_0 + \sum_{n=1}^{n=\infty} (a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t))$$

Parseval's theorem becomes

$$\frac{1}{T} \int_{\langle T \rangle} |x(t)|^2 dt = a_0^2 + \frac{1}{2} \sum_{n=1}^{n=\infty} (a_n^2 + b_n^2) \quad (39)$$

Physically this can be interpreted in terms of power being dissipated within a resistor, which can be represented by

$$P = \frac{V^2}{R}$$

for a perfect component. If a purely sinusoidal voltage of amplitude A was applied to a 1Ω resistor then over a single period the mean power dissipated would be

$$\begin{aligned} \frac{1}{2\pi} \int_{-\pi}^{\pi} (A \sin(nt))^2 dt &= \frac{A^2}{2\pi} \int_{-\pi}^{\pi} \sin^2(nt) dt \\ &= \frac{A^2}{2\pi} \int_{-\pi}^{\pi} \frac{1}{2} (1 - \cos(2nt)) dt = \frac{A^2}{4\pi} \left[t - \frac{1}{n} \sin(2nt) \right]_{-\pi}^{\pi} = \frac{A^2}{2} \end{aligned}$$

A similar result holds for a cosine signal. Parseval's theorem hence states that the total power dissipation is the sum of the power dissipated by each separate sinusoidal frequency component.

In terms of the amplitude-phase form of series, the theorem becomes

$$\frac{1}{T} \int_{\langle T \rangle} |x(t)|^2 dt = a_0^2 + \frac{1}{2} \sum_{n=1}^{n=\infty} A_n^2 \quad (40)$$

2.7. SIGNAL SHIFTS AND SYMMETRIES

It has already been noted that if a function is odd over the interval of integration, then the corresponding Fourier expansion will only involve pure sine terms. Likewise, an even series can be expressed purely in terms of cosine terms plus a constant DC offset.

Other symmetries and structure can also lead to specific patterns within a Fourier series expansion.

Odd half-wave symmetry

The concept of *odd half-wave symmetry* is applicable to periodic signals and describes the case where, when examining a single period of the signal the values of second half of the signal period are merely the negative of those of the first half. In detail:

$$x\left(t\right)=-x\left(t+\frac{T}{2}\right) \quad (41)$$

so that shifting the second half of the signal through one-half of a period and adding it to the first half will produce a zero result.

As illustrated below, a function can be neither even, nor odd, but can still display odd half-wave symmetry.

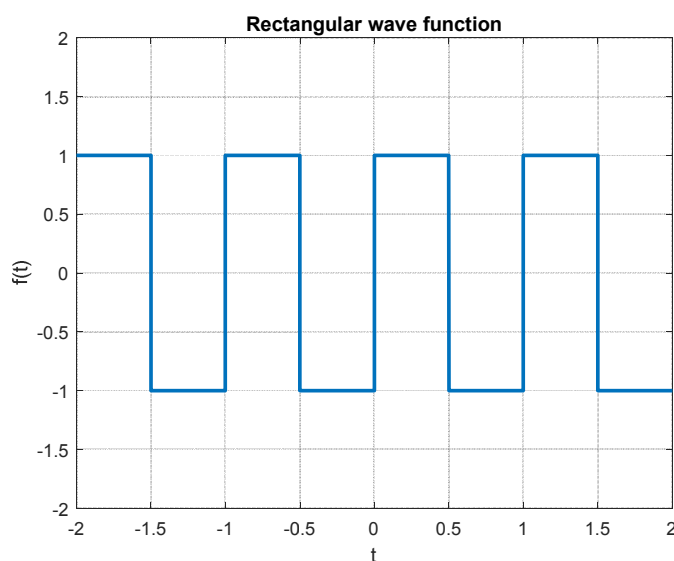


Figure 12 An odd function which displays half-wave symmetry (see Example 3).

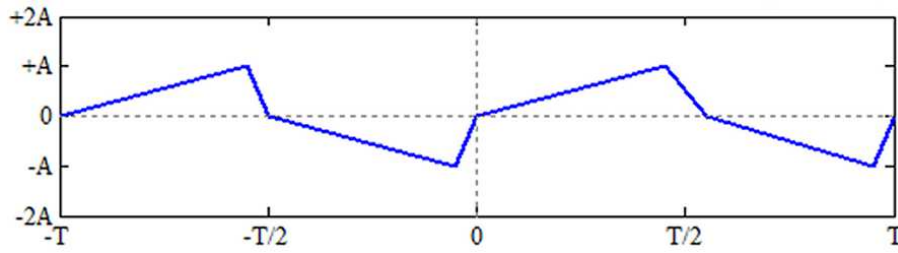


Figure 13 A signal of period T which is neither even nor odd, but which displays odd half-wave symmetry.

If a function does satisfy

$$x(t) = -x\left(t + \frac{T}{2}\right)$$

when evaluating the Fourier coefficients we can break the integration into two parts, so that:

$$\begin{aligned} a_n &= \frac{2}{T} \int_{-T/2}^{T/2} x(t) \cos(n\omega_0 t) dt \\ &= \frac{2}{T} \int_{-T/2}^0 x(t) \cos(n\omega_0 t) dt + \frac{2}{T} \int_0^{T/2} x(t) \cos(n\omega_0 t) dt \\ &= \frac{2}{T} \int_{-T/2}^0 x(t) \cos(n\omega_0 t) dt + \frac{2}{T} \int_{-T/2}^0 x\left(u + \frac{T}{2}\right) \cos\left(n\omega_0 \left(u + \frac{T}{2}\right)\right) du \\ &= \frac{2}{T} \int_{-T/2}^0 x(t) \cos(n\omega_0 t) dt - \frac{2}{T} \int_{-T/2}^0 x(u) \cos(n\omega_0 u + n\pi) du \\ &= \frac{2}{T} \int_{-T/2}^0 x(t) \cos(n\omega_0 t) dt - \frac{2}{T} \int_{-T/2}^0 x(u) \cos(n\omega_0 u) \cos(n\pi) du \\ &= \frac{2}{T} \int_{-T/2}^0 x(t) \cos(n\omega_0 t) dt - \frac{2}{T} \int_{-T/2}^0 x(u) \cos(n\omega_0 u) (-1)^n du \end{aligned}$$

Which vanishes for any even value of n . Similarly

$$\begin{aligned} b_n &= \frac{2}{T} \int_{-T/2}^{T/2} x(t) \sin(n\omega_0 t) dt \\ &= \frac{2}{T} \int_{-T/2}^0 x(t) \sin(n\omega_0 t) dt + \frac{2}{T} \int_0^{T/2} x(t) \sin(n\omega_0 t) dt \\ &= \frac{2}{T} \int_{-T/2}^0 x(t) \sin(n\omega_0 t) dt + \frac{2}{T} \int_{-T/2}^0 x\left(u + \frac{T}{2}\right) \sin\left(n\omega_0 \left(u + \frac{T}{2}\right)\right) du \\ &= \frac{2}{T} \int_{-T/2}^0 x(t) \sin(n\omega_0 t) dt - \frac{2}{T} \int_{-T/2}^0 x(u) \sin(n\omega_0 u + n\pi) du \\ &= \frac{2}{T} \int_{-T/2}^0 x(t) \sin(n\omega_0 t) dt - \frac{2}{T} \int_{-T/2}^0 x(u) \sin(n\omega_0 u) \cos(n\pi) du \\ &= \frac{2}{T} \int_{-T/2}^0 x(t) \sin(n\omega_0 t) dt - \frac{2}{T} \int_{-T/2}^0 x(u) \sin(n\omega_0 u) (-1)^n du \end{aligned}$$

which also vanishes for any even value of n .

Finally, the constant term must be zero by symmetry. Or, formally

$$\begin{aligned}
 a_0 &= \frac{1}{T} \int_{-T/2}^{T/2} x(t) dt \\
 &= \frac{1}{T} \int_{-T/2}^0 x(t) dt + \frac{1}{T} \int_0^{T/2} x(t) dt \\
 &= \frac{1}{T} \int_{-T/2}^0 x(t) dt + \frac{1}{T} \int_{-T/2}^0 x\left(u + \frac{T}{2}\right) du \\
 &= \frac{1}{T} \int_{-T/2}^0 x(t) dt - \frac{1}{T} \int_{-T/2}^0 x(u) du = 0
 \end{aligned}$$

Overall:

- A function which displays odd half-wave symmetry can be constructed from only odd harmonics.

Quarter-wave symmetry

If a signal is half-wave symmetric and also has odd or even symmetry about a quarter-period point (a distance of $\frac{T}{4}$ from an end, or the centre), then it is said to be *quarter-wave symmetric*.

Even quarter-wave symmetry is when

$$x(t) = -x\left(t + \frac{T}{2}\right)$$

and

$$x(-t) = x(t) \quad (42)$$

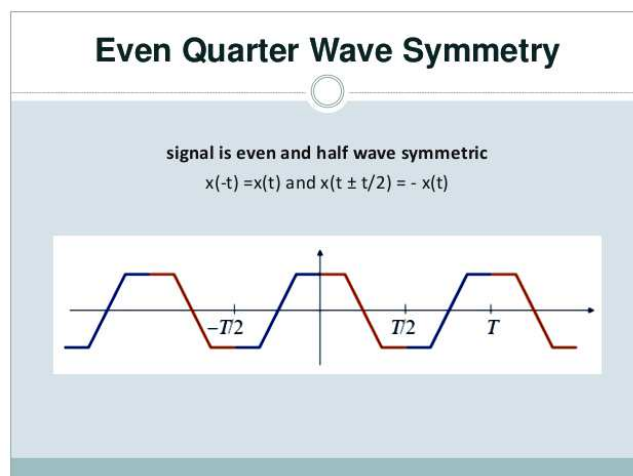


Figure 14 A signal displaying even quarter-wave symmetry.

Overall:

- A function which displays even quarter-wave symmetry can be constructed from only odd cosine harmonics.

Odd quarter-wave symmetry is when

$$x(t) = -x\left(t + \frac{T}{2}\right)$$

and

$$x(-t) = -x(t) \quad (43)$$

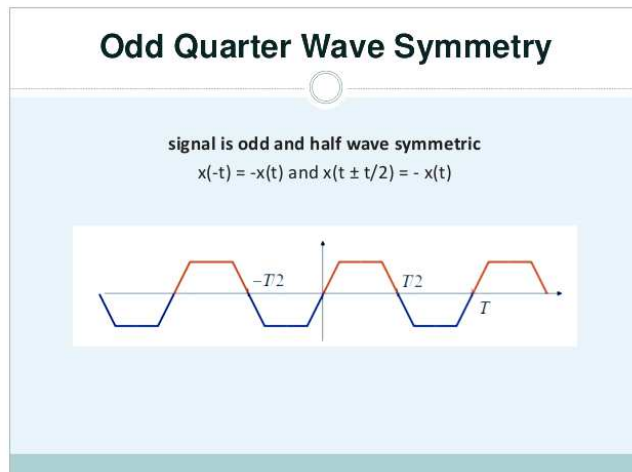


Figure 15 A signal displaying odd quarter-wave symmetry.

Overall:

- A function which displays odd quarter-wave symmetry can be constructed from only odd sine harmonics.

As before, a periodic function can be even or odd over an interval depending on the chosen position of the interval along the time axis; the exact Fourier series expansion varies according to where the interval is placed along the periodic signal.

2.8. THE FOURIER TRANSFORM

Fourier series expansions are only appropriate for handling continuous-time periodic functions, and produce a discrete line spectrum with a fixed spacing of $\omega_0 = \frac{2\pi}{T}$ rad s⁻¹ (or $f_0 = \frac{1}{T}$ Hz) between the spectral lines. This means that functions with longer periods produce spectra where the spectral lines are closer together. If the period of the function is allowed to grow arbitrarily large, the spectral lines will move correspondingly closer together and, in the limit as $T \rightarrow \infty$, it can be envisaged that a Fourier analysis of a non-periodic signal would produce a continuous spectral description.

The expression

$$x(t) = \sum_{n=-\infty}^{n=\infty} c_n e^{jn\omega_0 t}$$

define the complex exponential form of a Fourier series, with the coefficients obtained from

$$c_n = \frac{1}{T} \int_{-T/2}^{T/2} x(t) \exp(-jn\omega_0 t) dt$$

As T increases, $\omega_0 = \frac{2\pi}{T}$ would become arbitrarily small, say $\Delta\omega$, and combining the equations above gives

$$\begin{aligned} x(t) &= \sum_{n=-\infty}^{n=\infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) \exp(-jn\omega_0 t) dt e^{jn\omega_0 t} \\ &= \frac{1}{2\pi} \sum_{n=-\infty}^{n=\infty} \int_{-T/2}^{T/2} x(\tau) \exp(-jn\omega_0 \tau) d\tau e^{jn\omega_0 t} \Delta\omega \\ &= \frac{1}{2\pi} \sum_{n=-\infty}^{n=\infty} e^{jn\omega_0 t} \int_{-T/2}^{T/2} x(\tau) \exp(-jn\omega_0 \tau) d\tau \Delta\omega \end{aligned}$$

using a dummy variable of integration.

As $T \rightarrow \infty$ the previously discrete angular frequencies $n\omega_0 = \frac{2\pi n}{T}$ become a continuous variable, ω , and the infinite sum becomes an integral:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{j\omega t} \int_{-\infty}^{\infty} x(\tau) \exp(-j\omega\tau) d\tau d\omega$$

This leads to the definition of the *Fourier transform*:

$$\mathfrak{F}[x(t)] = X(\omega) = \hat{x}(\omega) \triangleq \int_{-\infty}^{\infty} x(t) \exp(-j\omega t) dt \quad (44)$$

which changes the viewpoint from that of an infinite, non-periodic continuous-time signal $x(t)$ to an infinite non-periodic, continuous-frequency spectrum $X(\omega)$.

The inverse Fourier transform

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \quad (45)$$

reverses the process and switches from frequency back to time.

The Fourier transform is the appropriate tool to use when the signal $x(t)$ is single-valued, aperiodic and is a function of a continuous-time variable t . As with Fourier series expansions, there are mathematical criteria for a Fourier transform to exist, but the dominant one is that the signal $x(t)$ must be absolutely integrable - that is

$$\|x\|_1 = \int_{-\infty}^{\infty} |x(t)| dt < \infty$$

$\|x\|_1$ is called the *L1 norm* and is one measure of the total size of the signal.

2.9. PROPERTIES OF THE FOURIER TRANSFORM

The Fourier transform appears to be very similar to the *two-sided Laplace transform*, defined by:

$$F(s) = L[x(t)] \triangleq \int_{-\infty}^{\infty} x(t) e^{-st} dt \quad (46)$$

where s is a complex variable.

For many applications in signal processing and control theory it is assumed that $f(t) = 0$ for $t < 0$ and the one-sided Laplace transform is used. But this doesn't have to be the case - there are other applications where the signal may need to be allowed to be non-zero for negative time. The two-sided Laplace transform is appropriate for such cases and, in effect, the Fourier transform can be obtained from the two-sided Laplace transform simply by setting $s = j\omega$. This means that many of the basic properties of the Fourier transform mirror those of the Laplace transform, although there are some differences.

Linearity:

$$\mathfrak{S}[ax(t) + by(t)] = a\mathfrak{S}[x(t)] + b\mathfrak{S}[y(t)] = aX(\omega) + bY(\omega)$$

Scaling:

$$\mathfrak{S}[x(at)] = \frac{1}{|a|} X\left(\frac{\omega}{a}\right) \quad \text{if } a \neq 0$$

Time differentiation:

$$\mathfrak{S}\left[\frac{dx}{dt}\right] = j\omega X(\omega) = j2\pi X(f)$$

Frequency differentiation:

$$\mathfrak{S}[tx(t)] = j \frac{dX}{d\omega}$$

First shifting theorem (frequency shifting):

$$\mathfrak{S}[e^{j\omega_0 t} x(t)] = X(\omega - \omega_0)$$

Second shifting theorem (time shifting):

$$\mathfrak{S}[x(t - t_0)] = e^{-j\omega t_0} X(\omega)$$

Time integration:

$$\mathfrak{S}\left[\int_{-\infty}^t x(\tau) d\tau\right] = \frac{X(\omega)}{j\omega} + c\delta(\omega) \quad \text{where } \int_{-\infty}^{\infty} (x(t) - c) dt = 0$$

Time convolution (frequency modulation):

$$\mathfrak{S}[x(t) * y(t)] = \mathfrak{S}\left[\int_{-\infty}^{\infty} x(\tau) y(t - \tau) d\tau\right] = X(\omega) Y(\omega)$$

Time modulation (frequency convolution):

$$\mathfrak{S}[x(t)y(t)] = X(\omega) * Y(\omega)$$

Duality:

$$\mathfrak{S}[X(t)] = 2\pi x(-\omega)$$

Time reversal:

$$\mathfrak{S}[x(-t)] = X(-\omega)$$

Parseval's theorem:

$$E_{Total} = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

which, as for line spectra, is a powerful result which effectively states that the total signal energy is the same whether the signal is viewed in the time or frequency domains. It also can be viewed as representing the fact that information is not lost by carrying out the Fourier transform or its inverse.

The $L2$ norm is $\|x\|_2$, defined by

$$\|x\|_2^2 = \int_{-\infty}^{\infty} |x(t)|^2 dt$$

and is used in Parseval's theorem where it represents the signal energy. An *energy signal* is one where the total energy is finite:

$$E_{Total} = \int_{-\infty}^{\infty} |x(t)|^2 dt < \infty$$

Some texts would state that such a signal is *square integrable*. Periodic functions do not have finite total energy and are called *power signals*.

The various properties can be proved directly from the definition of the Fourier transform given in equation (44).

Differentiation

What happens if we wish to evaluate the Fourier transform of the time derivative of a given signal: $\mathfrak{F}\left[\frac{dx}{dt}\right]$?

From the definition

$$\mathfrak{F}\left[\frac{dx}{dt}\right] = \int_{-\infty}^{\infty} \frac{dx}{dt} \exp(-j\omega t) dt$$

Use integration by parts, and let $u = e^{-j\omega t}$ and $\frac{dv}{dt} = \frac{dx}{dt}$. Then

$$\begin{aligned} \mathfrak{F}\left[\frac{dx}{dt}\right] &= \int_{-\infty}^{\infty} \exp(-j\omega t) \frac{dx}{dt} dt \\ &= \left[e^{-j\omega t} x(t) \right]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} (-j\omega e^{-j\omega t}) x(t) dt \\ &= [0 - 0] + j\omega \int_{-\infty}^{\infty} x(t) e^{-j\omega t} dt = j\omega X(\omega) \end{aligned}$$

where it is being assumed that the function dies away to zero as $t \rightarrow \infty$. This is perfectly reasonable if the signal has finite energy.

This result parallels the Laplace transform result that

$$L\left[\frac{df}{dt}\right] = sF(s) - f(0)$$

with the difference that there is no initial condition, since the signal is assumed to be arbitrarily small as $t \rightarrow -\infty$.

Time integration:

From the definition of the Fourier transform

$$\begin{aligned}\mathfrak{F}\left[\int_{-\infty}^t x(\tau) d\tau\right] &= \int_{-\infty}^{\infty} \int_{-\infty}^t x(\tau) d\tau \exp(-j\omega t) dt \\ &= \int_{-\infty}^{\infty} x(\tau) \int_{\tau}^{\infty} \exp(-j\omega t) dt d\tau \\ &= \int_{-\infty}^{\infty} x(\tau) \left[-\frac{1}{j\omega} \exp(-j\omega t)\right]_{\tau}^{\infty} d\tau \\ &= \int_{-\infty}^{\infty} x(\tau) \left(0 + \frac{1}{j\omega} \exp(-j\omega\tau)\right) d\tau \\ &= \frac{1}{j\omega} \int_{-\infty}^{\infty} x(\tau) \exp(-j\omega\tau) d\tau = \frac{X(\omega)}{j\omega}\end{aligned}$$

except that this result is invalid when $\omega = 0$. This parallels the fact that in sinusoidal Fourier series, the a_0 term, corresponding to any constant DC component of the signal, may have to be handled separately to the a_n and b_n terms, which correspond to true sinusoids.

The trick to avoid this problem is to modify the original signal to remove any DC component by introducing an additive constant, c which satisfies

$$\int_{-\infty}^{\infty} (x(t) - c) dt = 0$$

For a new signal, $x(t) - c$, which has no $\omega = 0$ component, it is true to say that

$$\mathfrak{F}\left[\int_{-\infty}^t (x(\tau) - c) d\tau\right] = \frac{X(\omega)}{j\omega}$$

and for the original signal, the result can be modified to be

$$\mathfrak{F}\left[\int_{-\infty}^t x(\tau) d\tau\right] = \frac{X(\omega)}{j\omega} + c\delta(\omega)$$

where $\int_{-\infty}^{\infty} (x(t) - c) dt = 0$.

First shifting theorem (frequency shifting):

The Fourier transform of any signal $x(t)$ multiplied by a complex exponential $e^{j\omega_0 t}$ is, from the definition of the Fourier transform:

$$\begin{aligned}\mathfrak{F}\left[e^{j\omega_0 t} x(t)\right] &= \int_{-\infty}^{\infty} \exp(j\omega_0 t) x(t) \exp(-j\omega t) dt \\ &= \int_{-\infty}^{\infty} x(t) \exp(-j(\omega - \omega_0)t) dt \\ &= X(\omega - \omega_0)\end{aligned}$$

The Laplace version of this result was

$$L\left[e^{at} f(t)\right] = F(s - a)$$

Second shifting theorem (time shifting):

Similarly, shifting a signal in time gives

$$\begin{aligned}\mathfrak{F}\left[x(t - T)\right] &= \int_{-\infty}^{\infty} x(t - T) \exp(-j\omega t) dt \\ &= \int_{-\infty}^{\infty} x(\tau) \exp(-j\omega(\tau + T)) d\tau \\ &= \exp(-j\omega T) \int_{-\infty}^{\infty} x(\tau) \exp(-j\omega\tau) d\tau = e^{-j\omega T} X(\omega)\end{aligned}$$

using the substitution $\tau = t - T$.

The Laplace version of this result was

$$L\left[u(t - T) f(t - T)\right] = e^{-sT} F(s)$$

Time reversal

From the definition

$$\begin{aligned}\mathfrak{F}\left[x(-t)\right] &= \int_{-\infty}^{\infty} x(-t) \exp(-j\omega t) dt \\ &= -\int_{\infty}^{-\infty} x(u) \exp(+j\omega u) du \\ &= \int_{-\infty}^{\infty} x(u) \exp(-j(-\omega)u) du = X(-\omega)\end{aligned}$$

using the substitution $u = -t$.

Duality

Starting with the definition of the inverse Fourier transform

$$\begin{aligned}
 x(t) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \\
 \Rightarrow 2\pi x(t) &= \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega \\
 \Rightarrow 2\pi x(-t) &= \int_{-\infty}^{\infty} X(\omega) e^{-j\omega t} d\omega \\
 \Rightarrow 2\pi x(-\omega) &= \int_{-\infty}^{\infty} X(t) e^{-j\omega t} dt = \mathfrak{S}[X(t)]
 \end{aligned}$$

Hence

$$\mathfrak{S}[X(t)] = 2\pi x(-\omega)$$

This is an extremely useful result - if the Fourier transform of a signal is known to have a certain form, then the result can effectively be reversed - giving an additional known result.

Time convolution (frequency modulation):

The convolution of two real continuous-time signals is defined by

$$x(t) * y(t) = \int_{-\infty}^{\infty} x(\tau) y(t - \tau) d\tau$$

Hence the Fourier transform of this convolution integral is:

$$\begin{aligned}
 \mathfrak{S}[x(t) * y(t)] &= \mathfrak{S}\left[\int_{-\infty}^{\infty} x(\tau) y(t - \tau) d\tau\right] \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x(\tau) y(t - \tau) d\tau \exp(-j\omega t) dt
 \end{aligned}$$

This is a double integral involving two time variables, t and τ and the region of integration is the entire plane defined by the t and τ axes. Hence reversing the order of integration does not change the limits:

$$\begin{aligned}
 \mathfrak{S}[x(t) * y(t)] &= \mathfrak{S}\left[\int_{-\infty}^{\infty} x(\tau) y(t - \tau) d\tau\right] \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x(\tau) y(t - \tau) d\tau \exp(-j\omega t) dt \\
 &= \int_{-\infty}^{\infty} x(\tau) \int_{-\infty}^{\infty} y(t - \tau) \exp(-j\omega t) dt d\tau
 \end{aligned}$$

Also, using the substitution $u = t - \tau$:

$$\begin{aligned}
\mathfrak{F}[x(t) * y(t)] &= \int_{-\infty}^{\infty} x(\tau) \int_{-\infty}^{\infty} y(u) \exp(-j\omega(u + \tau)) dt d\tau \\
&= \int_{-\infty}^{\infty} x(\tau) \exp(-j\omega\tau) \int_{-\infty}^{\infty} y(u) \exp(-j\omega u) dt d\tau \\
&= \int_{-\infty}^{\infty} x(\tau) \exp(-j\omega\tau) Y(\omega) d\tau \\
&= Y(\omega) \int_{-\infty}^{\infty} x(\tau) \exp(-j\omega\tau) d\tau = X(\omega) Y(\omega)
\end{aligned}$$

Again, this parallels the Laplace convolution result that:

$$L\left[\int_0^t h(t - \tau) x(\tau) d\tau\right] = H(s) X(s)$$

2.10. THE FOURIER TRANSFORM OF SPECIFIC SIGNALS

Fourier transform of a unit impulse (Dirac delta function)

From the definition

$$\mathfrak{F}[\delta(t)] = \int_{-\infty}^{\infty} \delta(t) \exp(-j\omega t) dt = 1$$

which is identical to the Laplace result that

$$L[\delta(t)] = 1$$

Fourier transform of a (Heavyside) unit step function

As with the unit impulse, the unit step is a mathematical special case - it has a Fourier transform, even though it does not satisfy the requirement that $\|u(t)\|_1 = \int_{-\infty}^{\infty} |u(t)| dt < \infty$. Since the unit step function can be obtained from the unit impulse by integration, the time integration result can be used to give

$$\mathfrak{F}[u(t)] = \frac{1}{j\omega} + \frac{1}{2} \delta(\omega)$$

which is similar to the Laplace result that

$$L[u(t)] = \frac{1}{s}$$

The difference is the additional term which represents the fact that the unit step function has an average value of 0.5 when averaged over the range $-\infty < t < \infty$.

Fourier transform of a constant function

A constant function $x(t) = 1$ can be constructed as the sum of a step function and a time reversed step function, so

$$\mathfrak{F}[u(t) + u(-t)] = \left(\frac{1}{j\omega} + \frac{1}{2} \delta(\omega) \right) + \left(-\frac{1}{j\omega} + \frac{1}{2} \delta(\omega) \right) = \delta(\omega)$$

Note that this also satisfies the duality principle in the sense that the Fourier transform of a time domain impulse is a constant in the frequency domain, while the Fourier transform of a constant in the time domain is an impulse in the frequency domain.

Fourier transform of a rectangular pulse

In general the rectangular pulse $\text{rect}\left(\frac{t-T}{\tau}\right)$ is a function with unit height, width τ and is centred around the point $t = T$.

$$\text{In particular} \quad \text{rect}\left(\frac{t}{\tau}\right) = \begin{cases} 1 & \text{if } -\frac{\tau}{2} < t < \frac{\tau}{2} \\ 0 & \text{otherwise} \end{cases}$$

Hence, from the definition:

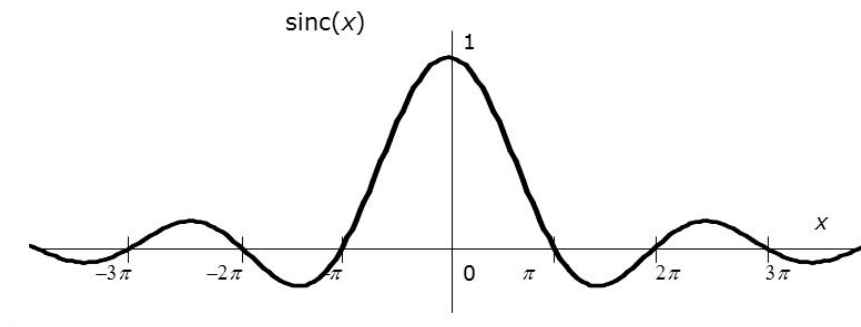
$$\begin{aligned} \mathfrak{F}\left[\text{rect}\left(\frac{t}{\tau}\right)\right] &= \int_{-\tau/2}^{\tau/2} \exp(-j\omega t) dt = \left[-\frac{1}{j\omega} \exp(-j\omega t) \right]_{-\tau/2}^{\tau/2} \\ &= -\frac{1}{j\omega} \left(\exp\left(-j\omega \frac{\tau}{2}\right) - \exp\left(j\omega \frac{\tau}{2}\right) \right) \\ &= \frac{2}{\omega} \sin\left(\frac{\omega\tau}{2}\right) = \tau \text{sinc}\left(\frac{\omega\tau}{2}\right) \end{aligned}$$

OR Build the pulse as the difference between two shifted unit steps:

$$\begin{aligned} \mathfrak{F}\left[\text{rect}\left(\frac{t}{\tau}\right)\right] &= \mathfrak{F}\left[u\left(t + \frac{\tau}{2}\right) - u\left(t - \frac{\tau}{2}\right)\right] \\ &= \left(\frac{1}{j\omega} \exp\left(j\omega \frac{\tau}{2}\right) + \frac{1}{2} \delta(\omega) \right) - \left(\frac{1}{j\omega} \exp\left(-j\omega \frac{\tau}{2}\right) + \frac{1}{2} \delta(\omega) \right) \\ &= \frac{1}{j\omega} \left(\exp\left(j\omega \frac{\tau}{2}\right) - \exp\left(-j\omega \frac{\tau}{2}\right) \right) \\ &= \frac{2}{\omega} \sin\left(\frac{\omega\tau}{2}\right) = \tau \text{sinc}\left(\frac{\omega\tau}{2}\right) \end{aligned}$$

The function $\text{rect}\left(\frac{t}{\tau}\right)$ is a model for a perfect low-pass filter and so

$\text{sinc}(x) = \frac{\sin(x)}{x}$ arises a great deal in signal processing.



Note that $\lim_{x \rightarrow 0} [\text{sinc}(x)] = 1$

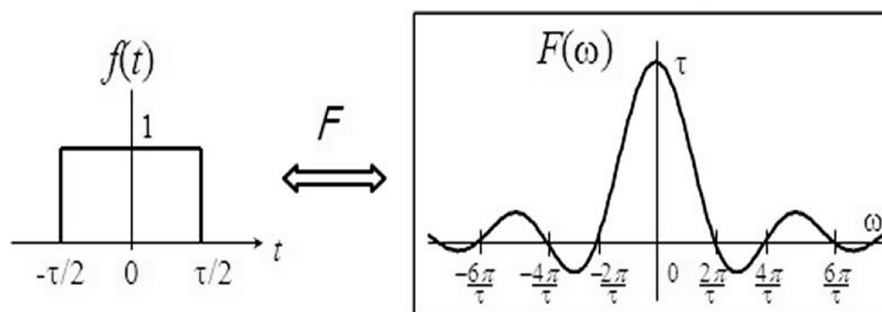


Figure 16 A rectangular pulse $\text{rect}\left(\frac{t}{T}\right)$
and its Fourier transform $\tau \text{sinc}\left(\frac{\omega\tau}{2}\right)$.

Fourier transform of a complex exponential

Since $\mathfrak{F}[1] = \delta(\omega)$ we can use the frequency shifting theorem to give the Fourier transform of a complex exponential as:

$$\mathfrak{F}[\exp(j\omega_0 t)] = \delta(\omega - \omega_0)$$

Fourier transform of a cosine signal

We can use the complex exponential form of $\cos(t)$ together with the previous result for the Fourier transform of a complex exponential:

$$\begin{aligned}\mathfrak{F}[\cos(\omega_0 t)] &= \frac{1}{2}(\mathfrak{F}[\exp(j\omega_0 t) + \exp(-j\omega_0 t)]) \\ &= \frac{1}{2}(\mathfrak{F}[\exp(j\omega_0 t)] + \mathfrak{F}[\exp(-j\omega_0 t)]) \\ &= \frac{1}{2}(\delta(\omega - \omega_0) + \delta(\omega + \omega_0))\end{aligned}$$

Fourier transform of a sine signal

Similarly

$$\begin{aligned}\mathfrak{F}[\sin(\omega_0 t)] &= \frac{1}{2j}(\mathfrak{F}[\exp(j\omega_0 t) - \exp(-j\omega_0 t)]) \\ &= \frac{1}{2j}(\mathfrak{F}[\exp(j\omega_0 t)] - \mathfrak{F}[\exp(-j\omega_0 t)]) \\ &= \frac{1}{2j}(\delta(\omega - \omega_0) - \delta(\omega + \omega_0))\end{aligned}$$

In general the Fourier transform is a complex function but as in the case of the cosine and sine examples above, it can be shown that:

- The Fourier transform of an even integrable function is an even real-valued function.
- The Fourier transform of an odd integrable function is an odd, imaginary function

Sinusoidal modulation

If a signal $x(t)$ is modulated (multiplied by) a cosine signal its Fourier transform will be

$$\begin{aligned}\mathfrak{F}[x(t) \cos(\omega_0 t)] &= \frac{1}{2}(\mathfrak{F}[x(t) \exp(j\omega_0 t) + x(t) \exp(-j\omega_0 t)]) \\ &= \frac{1}{2}(\mathfrak{F}[x(t) \exp(j\omega_0 t)] + \mathfrak{F}[x(t) \exp(-j\omega_0 t)]) \\ &= \frac{1}{2}(X(\omega - \omega_0) + X(\omega + \omega_0))\end{aligned}$$

using the First shifting theorem result that:

$$\mathfrak{F}\left[e^{j\omega_0 t} x(t)\right] = X(\omega - \omega_0)$$

Likewise

$$\begin{aligned}\mathfrak{F}\left[x(t) \sin(\omega_0 t)\right] &= \frac{1}{2j} \left(\mathfrak{F}\left[x(t) \exp(j\omega_0 t)\right] - \mathfrak{F}\left[x(t) \exp(-j\omega_0 t)\right] \right) \\ &= \frac{1}{2j} \left(\mathfrak{F}\left[x(t) \exp(j\omega_0 t)\right] - \mathfrak{F}\left[x(t) \exp(-j\omega_0 t)\right] \right) \\ &= \frac{1}{2j} \left(X(\omega - \omega_0) - X(\omega + \omega_0) \right)\end{aligned}$$

This frequency-shifting effect is seen in practice with *amplitude modulated* (AM) signals, where a high-frequency (*radio-frequency*, RF) sinusoidal *carrier wave* is modulated by a lower-frequency information signal which is to be transmitted. The result is illustrated below.

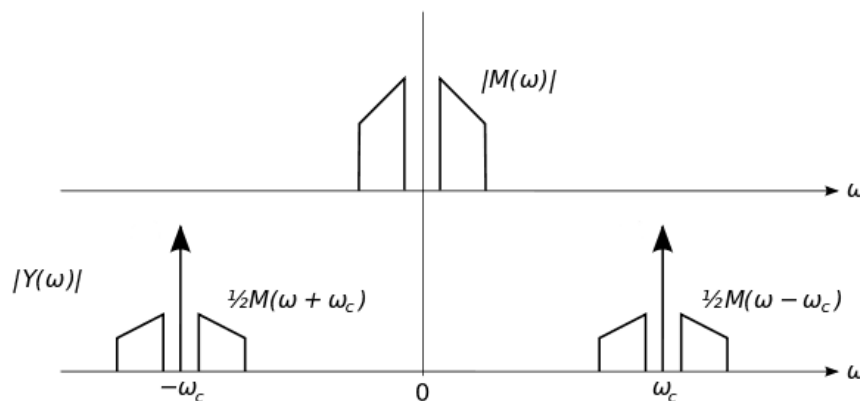


Figure 17 The effect of modulating a RF sinusoidal carrier wave of frequency ω_c with an information signal $m(t)$.

The figure shows that the *baseband* spectrum of the low-frequency information signal is shifted to form *sidebands* centred around the carrier-wave frequency ω_c .

The amplitude modulation approach was the first to be used for radio transmission in the early 20th century and is still widely used today.

2.11. THE FOURIER SPECTRUM

In effect, Fourier methods decompose a function of time into its constituent frequencies - they switch the description of a function from the time domain to the frequency domain - and vice versa.

For the case of the Fourier series of a periodic spectrum, the frequencies form a discrete line spectrum, with the separation between the frequencies being $\omega_0 = \frac{2\pi}{T}$ so that, as the sampling period is increased, the spectral lines move closer together.

For the Fourier transform of an aperiodic signal, the period has effectively become infinite and the spectral lines have merged to form a *continuous spectrum*. This is a frequency-domain description of the signal, where the Fourier transform

$$X(\omega) = \int_{-\infty}^{\infty} x(t) \exp(-j\omega t) dt$$

defines the *complex Fourier spectrum*. The same information is contained in the (real) magnitude and phase spectra, defined by

- magnitude spectrum: $|X(\omega)| = \text{abs}(X(\omega))$
- phase spectrum: $\arg(X(\omega))$

The amplitude spectrum gives a direct measure of the relative amounts of different frequencies with a signal and is often plotted on a logarithmic scale, as illustrated.

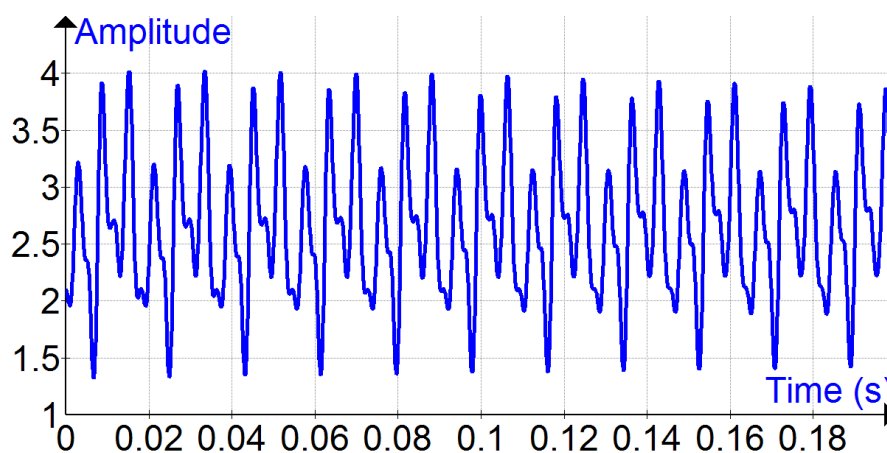


Figure 18 A time signal of open string bass guitar A note (55 Hz).
(from Wikipedia)

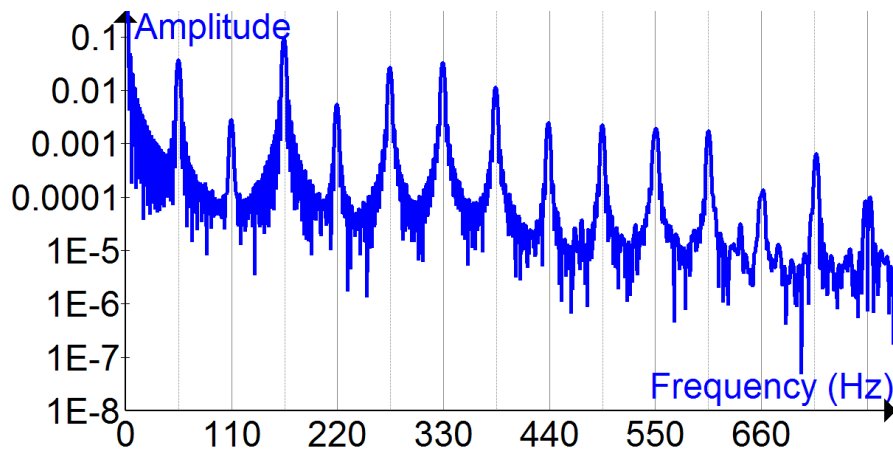


Figure 19 The amplitude (magnitude) of the Fourier transform of bass guitar time signal of open string A note (55 Hz).
(from Wikipedia)

These spectra not only allow visualization of information that might be unclear in the time-domain signal, but they also provide the basis for signal processing techniques such as filtering, equalization and noise reduction.

As with periodic functions, Parseval's theorem connects the total energy of the signal measured in the time domain with that in the frequency domain.

$$E_{Total} = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(\omega)|^2 d\omega$$

In this expression $|X(\omega)|^2$ effectively describes the energy per unit (angular) frequency contained in the signal as a function of ω - it is known as the *energy spectral density* of the aperiodic signal $x(t)$.

$$S_{xx}(\omega) = |X(\omega)|^2$$

Energy spectra are appropriate to signals whose energy is time-limited, but the concept of *power spectral density* (PSD) or *power spectrum* is used more often since it represents the energy spectral density per unit time, and hence can be applied to power signals that exist over all time.

2.12. CONVOLUTION

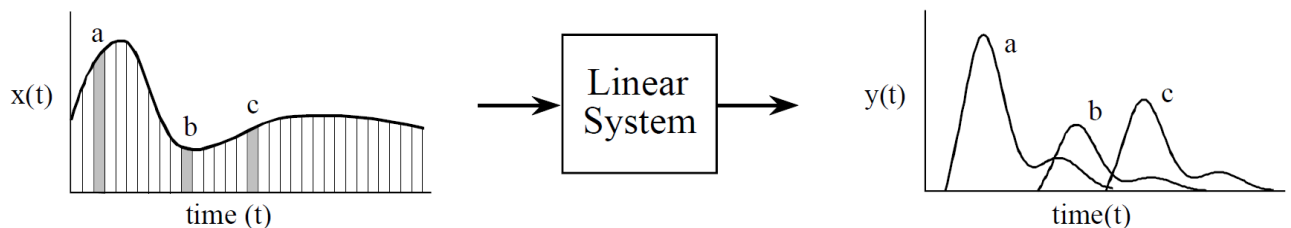
The convolution integral in the time domain is given by

$$y(t) = x * h = (x * y)(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau \quad (47)$$

and the convolution property of the Fourier transform is that the output of a signal $x(t)$ passing through a filter with impulse response $h(t)$ can be represented by or simply the product of the two Fourier transforms in the frequency domain:

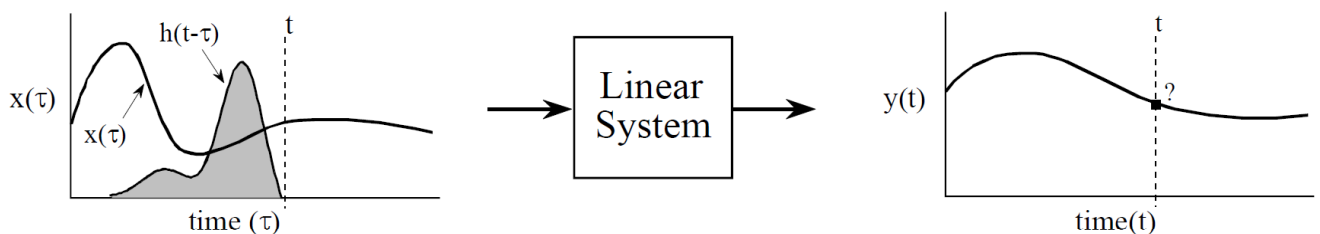
$$\mathfrak{F}[x(t) * h(t)] = \mathfrak{F}\left[\int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau\right] = X(\omega) H(\omega)$$

There are two useful ways to visualize this in the time domain:



Convolution viewed from the input side. The input signal, $x(t)$, is divided into narrow segments, each acting as an impulse to the system. The output signal, $y(t)$, is the sum of the resulting scaled and shifted impulse responses. This illustration shows how three points in the input signal contribute to the output signal.

Figure 20 Time-domain convolution viewed as a series of impulse responses.
(From "The Scientist and Engineer's Guide to Digital Signal Processing", By Steven W. Smith)

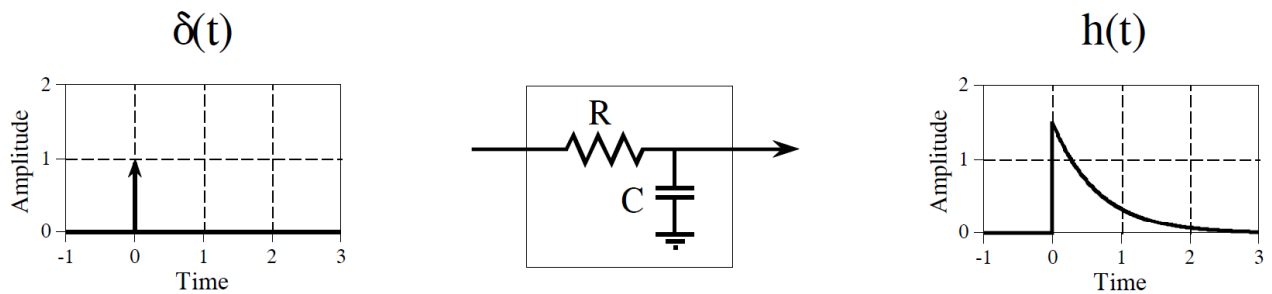


Convolution viewed from the output side. Each value in the output signal is influenced by many points from the input signal. In this figure, the output signal at time t is being calculated. The input signal, $x(\tau)$, is *weighted* (multiplied) by the flipped and shifted impulse response, given by $h(t - \tau)$. Integrating the weighted input signal produces the value of the output point, $y(t)$

Figure 21 Time-domain convolution viewed as multiple contributions to the output.
(From "The Scientist and Engineer's Guide to Digital Signal Processing", By Steven W. Smith)

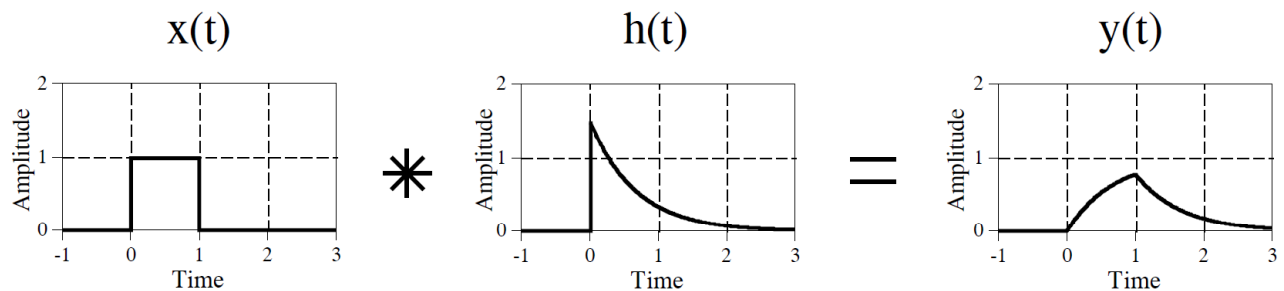
In practice, convolution arises in a massive number of engineering application areas. It represents the process of any signal passing through any system or, in a sense, of one signal being "blurred" by another.

Some basic examples of convolution are illustrated in Figure 22 and Figure 23:



Example of a continuous linear system. This electronic circuit is a low-pass filter composed of a single resistor and capacitor. The impulse response of this system is a one-sided exponential.

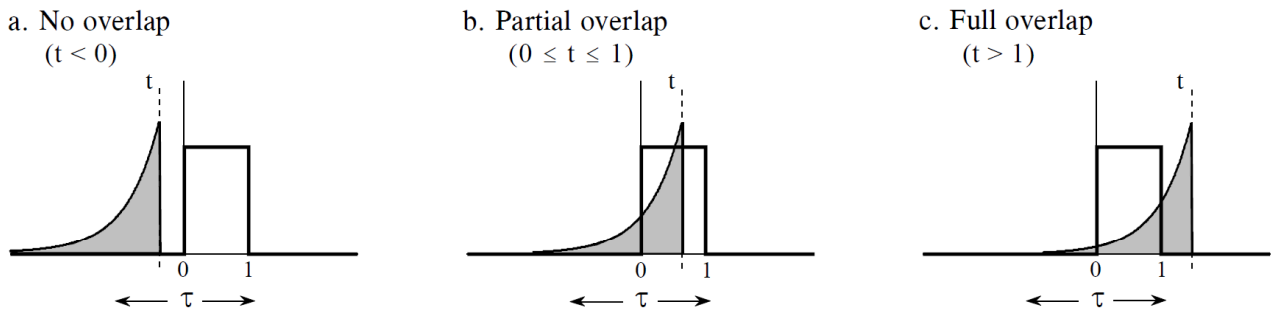
Figure 22 Time-domain convolution of an impulse through a RC circuit.
(From "The Scientist and Engineer's Guide to Digital Signal Processing", By Steven W. Smith)



Example of continuous convolution. This figure illustrates a square pulse entering an RC low-pass filter. The square pulse is convolved with the system's impulse response to produce the output.

Figure 23 Time-domain convolution of a rectangular pulse through a RC circuit.
(From "The Scientist and Engineer's Guide to Digital Signal Processing", By Steven W. Smith)

This latter example can be understood by viewing in more detail the multiplication and integration processes as the signal and the time-reversed impulse response are shifted to different relative positions in time.



Calculating a convolution by segments. Since many continuous signals are defined by *regions*, the convolution calculation must be performed region-by-region. In this example, calculation of the output signal is broken into three sections: (a) no overlap, (b) partial overlap, and (c) total overlap, of the input signal and the shifted-flipped impulse response.

Figure 24 Time-domain convolution of a rectangular pulse through a RC circuit.
(From "The Scientist and Engineer's Guide to Digital Signal Processing", By Steven W. Smith)

If the impulse response is given by $h(t) = A \exp(-\alpha t)$ then in the three regions illustrated, the convolution signal can be evaluated as three different integrals. In region (a), where $t < 0$

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t - \tau) d\tau = 0$$

since there is no overlap between the signals. In region (b), where $0 \leq t \leq 1$, the output is

$$\begin{aligned} y(t) &= \int_0^t A \exp(-\alpha(t - \tau)) d\tau \\ &= \left[\frac{A}{\alpha} \exp(-\alpha(t - \tau)) \right]_0^t = \frac{A}{\alpha} (1 - \exp(-\alpha t)) \end{aligned}$$

and in region (c), where $t > 1$ the output is

$$\begin{aligned} y(t) &= \int_0^1 A \exp(-\alpha(t - \tau)) d\tau \\ &= \left[\frac{A}{\alpha} \exp(-\alpha(t - \tau)) \right]_0^1 \\ &= \frac{A}{\alpha} (\exp(-\alpha(t - 1)) - \exp(-\alpha t)) \\ &= \frac{A(\exp(\alpha) - 1)}{\alpha} \exp(-\alpha t) \end{aligned}$$

giving the convolved output signal for all values of time.

Convolution is also defined for periodic signals, in which case

$$y(t) = x * h = (x * y)(t) = \frac{1}{T} \int_{\langle T \rangle} x(\tau) h(t - \tau) d\tau$$

and, similarly, finite limits can be used for time-limited signals.

2.13. CORRELATION

Cross-correlation

The process of establishing the *cross-correlation* of two (potentially complex) signals can be described by the integral definition:

$$x \star y = x \otimes y = x(t) \star y(t) = (x \star y)(t) = \int_{-\infty}^{\infty} \overline{x(\tau)} y(\tau + t) d\tau$$

or, for real signals,

$$x \star y = \int_{-\infty}^{\infty} x(\tau) y(\tau + t) d\tau \quad (48)$$

Except for the lack of time-reversal, this integral looks very like that of convolution and, mathematically, it is true that for real signals

$$x(t) \star y(t) = x(t) * y(-t)$$

Physically, they represent very different processes: convolution represents the outcome of a signal passing through a system with a specific impulse response, while cross-correlation provides a measure of the similarity of two signals.

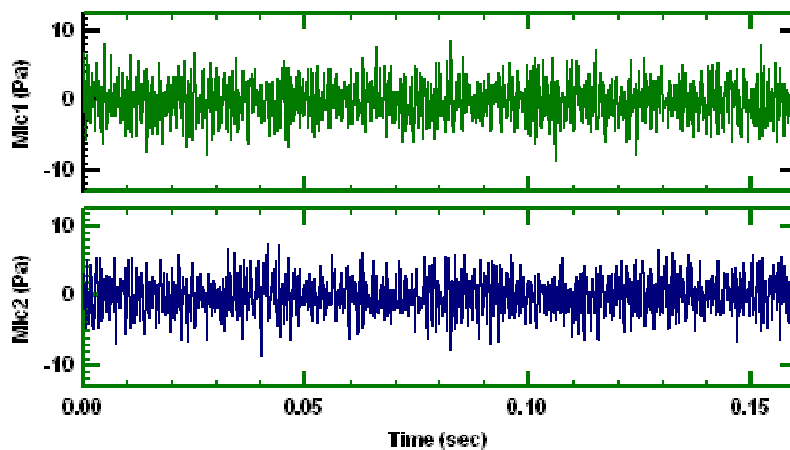


Figure 25 Two signals, corresponding to audio received at different times by two microphones, with each recording corrupted by different noise signals.

For example, Figure 25 shows two signals which have similar, but not identical content. Evaluating the cross-correlation between the two signals produces Figure 26, which indicates that there is a strong similarity between the signals corresponding to a lag of approximately 11ms, corresponding to a path difference of approximately 3.77m (assuming that the speed of sound is 343ms^{-1}).

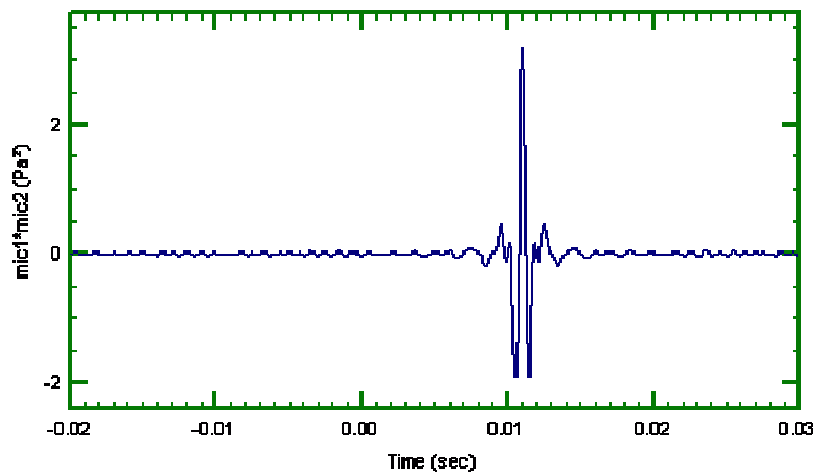


Figure 26 The cross-correlation of the signals in Figure 25.

When dealing with realistic, time-limited, data, one common approach is simply to assume that the signal is zero outside of the interval in question, as has been done above. Another is to create an artificially periodic version of the data, a process called using *periodic boundary conditions*. Similarly to convolution, cross-correlation is defined for periodic signals as

$$x \star y = \frac{1}{T} \int_{\langle T \rangle} x(\tau) y(\tau + t) d\tau$$

and, as before, finite limits can be used for time-limited signals.

The concepts of convolution and correlation can be extended to two, or more, dimensions and are used heavily in image processing and pattern recognition applications.

A more visual, two-dimensional, example of cross-correlation is given in Figure 27, where the cross-correlation between a video frame and a "model" image of a player produces a clear maximum at the point where the player is standing.

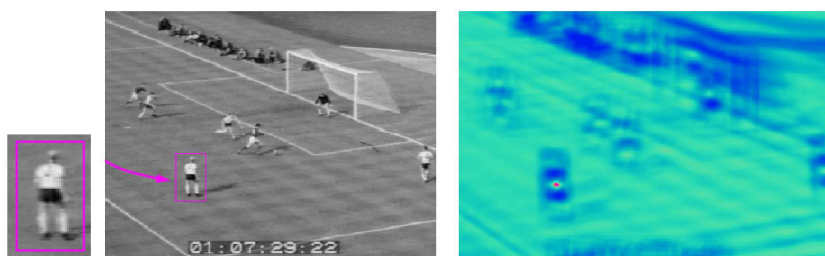


Figure 27 Cross-correlation applied to 2D image processing.

Figure 28, Figure 29 and Figure 30 show a further realistic example.

Figure 28 plots three signals representing measured neural activity from different neurons in the brain.

Figure 29 shows the three cross-correlations of the signals treated in pairs, with the large correlation peaks illustrating the fact that the signals are in fact very similar but are lagged relative to each other.

Figure 30 shows the three signals aligned after being adjusted for the relative delays identified by the cross-correlation results. Although the signals are clearly very similar, there are detailed differences between them.

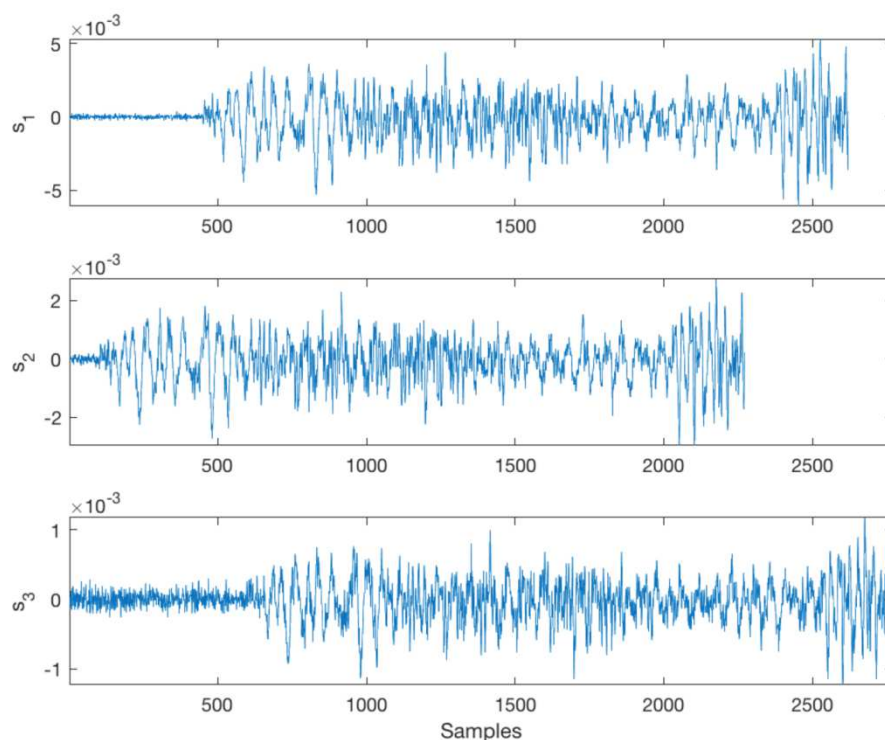


Figure 28 Three related neural signals
(source: <http://www.bradleymonk.com>).

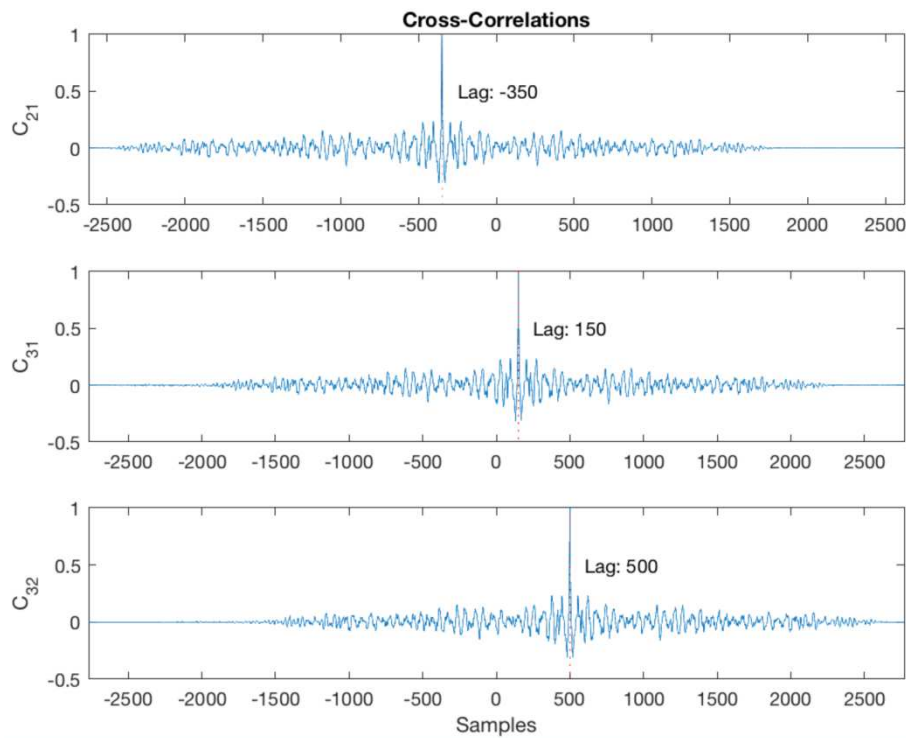


Figure 29 The cross-correlation plots for the three signals shown in Figure 28, treated in pairs.
(source: <http://www.bradleymonk.com>).

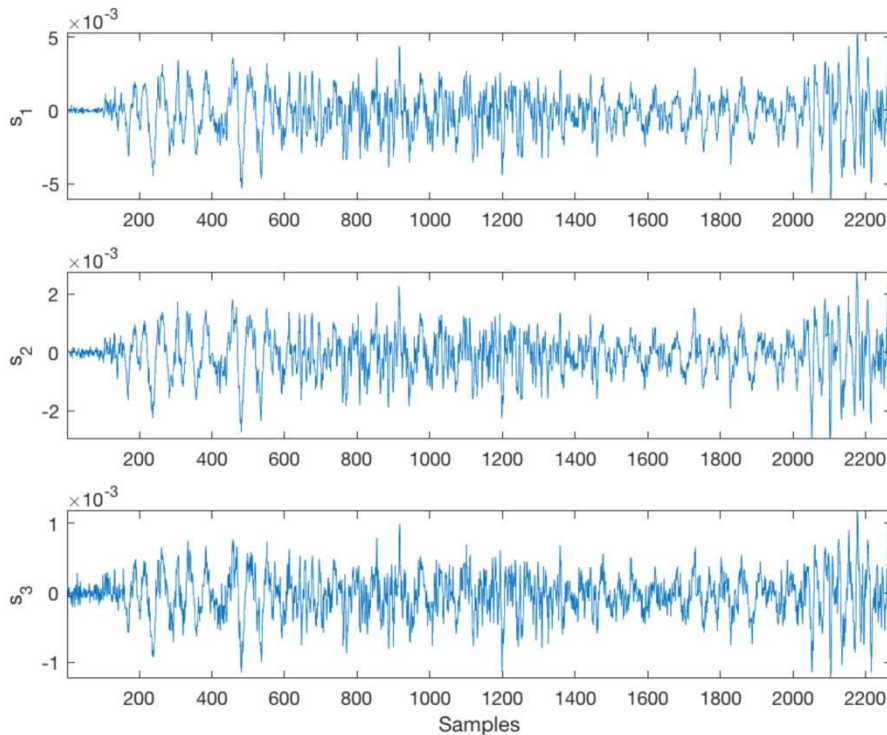


Figure 30 The neural signals after adjusting for the relative lags.
(source: <http://www.bradleymonk.com>).

Autocorrelation

The special case where the two signals are identical defines the autocorrelation of the signal:

$$x \star x = \int_{-\infty}^{\infty} x(\tau) x(\tau + t) d\tau \quad (49)$$

Again, in practice, the concept is often applied to short signals, and also to digital signals.

Autocorrelation can be used in a variety of applications, but is especially effective at finding periodic or repetitive signals in the presence of significant levels of noise. As well as being heavily used in science and engineering, it is a popular tool in the financial world for investors looking to find cycles or patterns in stock market prices!

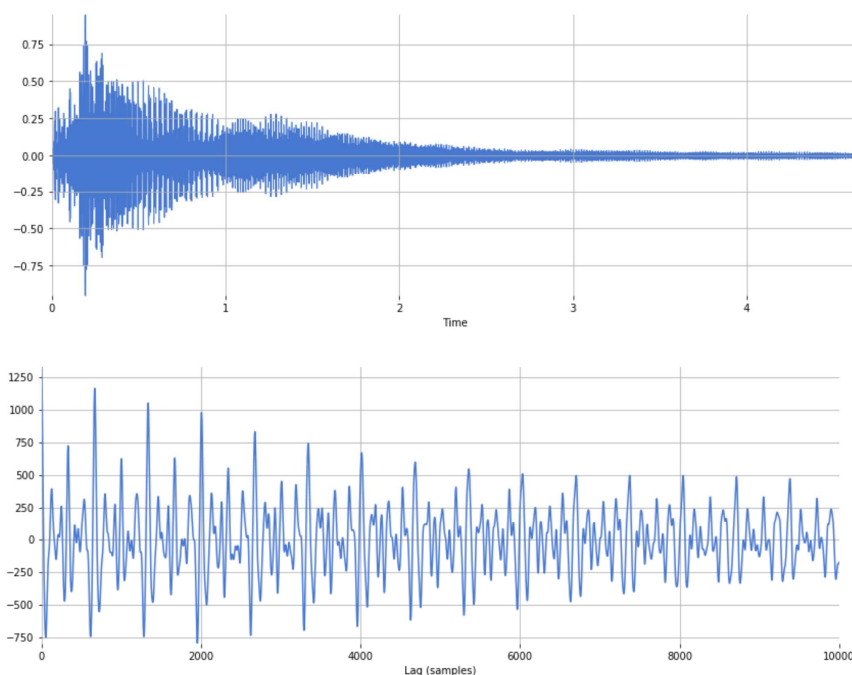


Figure 31 A guitar sound and its autocorrelation function.

Figure 31 shows a signal corresponding to a guitar sound, together with its autocorrelation function. The sound is sampled at 44.1kHz and so the series of strong peaks at lags of approximately 335 samples apart indicates a periodicity of $44100/335=131.6$ Hz, which is very close to the theoretical frequency of 130.81 Hz for the musical note C3. Figure 32 shows a simpler example of a noisy sine wave.

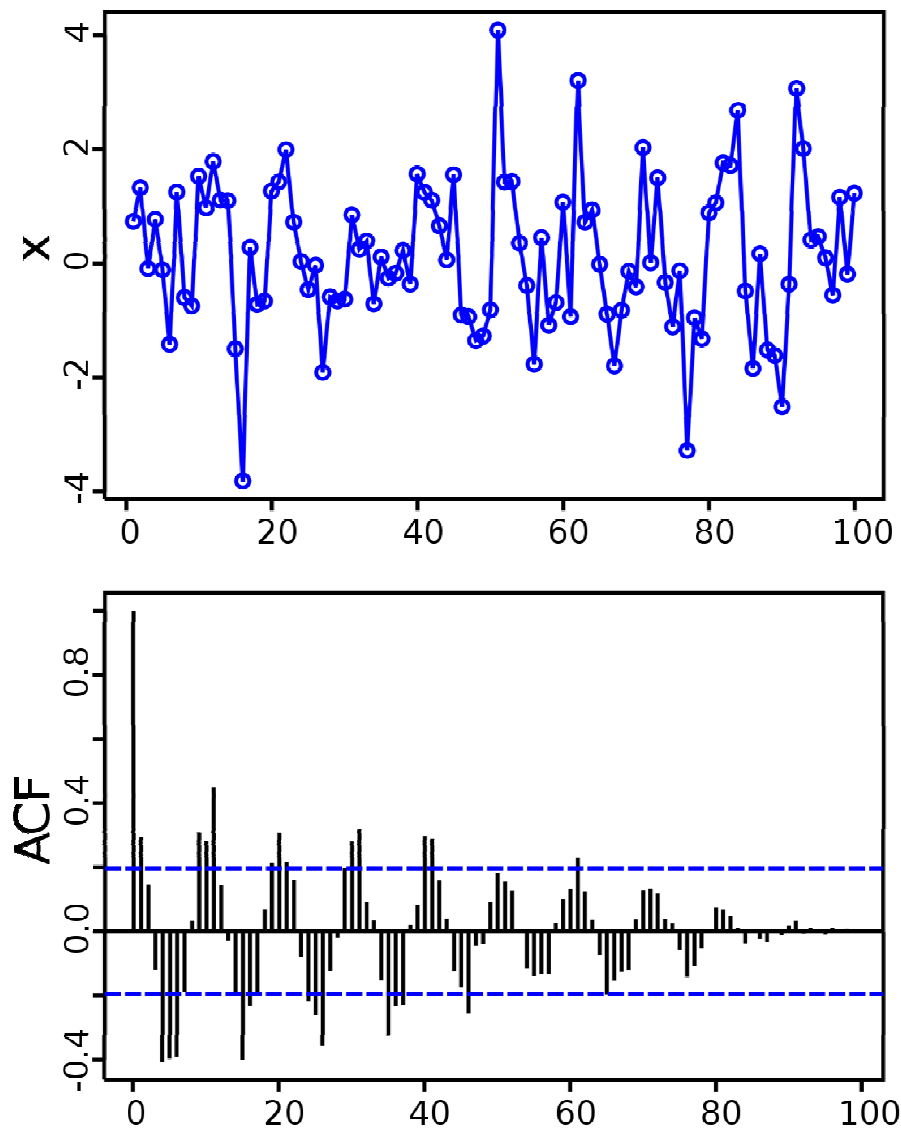


Figure 32 A sine wave, corrupted by random noise, and the autocorrelation of the signal, emphasising the periodicity of the signal (source: Wikipedia).

Note that the autocorrelation of a perfect white noise signal would be an impulse at zero lag and exactly zero for all other lags.

2.14. THE AUTOCORRELATION (WIENER-KHINCHIN) THEOREM

The autocorrelation (or Wiener-Khinchin) theorem states that the Fourier transform of the autocorrelation function is the spectral density.

$$\mathfrak{S}[x(t) \star x(t)] = \mathfrak{S}\left[\int_{-\infty}^{\infty} x(\tau) x(t + \tau) d\tau\right] = |X(\omega)|^2 \quad (50)$$

The result follows immediately from the convolution theorem:

$$\begin{aligned} \mathfrak{S}[x(t) \star y(t)] &= \mathfrak{S}\left[\int_{-\infty}^{\infty} x(\tau) y(t - \tau) d\tau\right] = X(\omega) Y(\omega) \\ \Rightarrow \mathfrak{S}[x(t) \star x(t)] &= \mathfrak{S}\left[\int_{-\infty}^{\infty} x(\tau) x(\tau + t) d\tau\right] \\ &= X(\omega) X(-\omega) = |X(\omega)|^2 \end{aligned}$$

using the fact that for real signals, $X(-\omega) = X(\omega)$. Only real signals are being considered here, but the result can be extended to complex signals.

For power signals the corresponding result gives the power spectral density (PSD):

$$\mathfrak{S}[x(t) \star x(t)] = \mathfrak{S}\left[\frac{1}{T} \int_{\langle T \rangle} x(\tau) x(\tau + t) d\tau\right] = \frac{1}{T} |X(\omega)|^2 \quad (51)$$